Undergraduate Specialization Project

Thermodynamic Design and Optimization of a Vapor-Compression Refrigeration Cycle with Ejector Expansion

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Abstract

In this project, a two-phase ejector-based vapor-compression refrigeration chiller is first investigated and analyzed. After the system has been thoroughly examined, the relevant thermodynamic equations are derived. The cycle is then implemented in MATLAB using the REFPROP add-on and fully simulated in order to determine its operating conditions. Finally, for specified condenser and evaporator temperatures, cycle outputs such as COP, VCC, and other performance parameters are obtained.

Next, a machine-learning algorithm is implemented to obtain a function that can rapidly predict COP for any specified temperature and intermediate-pressure input without repeatedly solving the governing cycle equations. By entering the required inputs, the user can readily obtain COP as well as other fluid properties at different state points of the cycle.

In the next step, the function obtained in the previous stage is used within an optimization algorithm to identify the maximum attainable COP of the cycle.

After completing the above calculations, the main design stage of the project—the ejector—is addressed. Using various references, an optimized configuration for the two-phase ejector is proposed.

It should be noted that, following initial validation, the project was planned to proceed to fabrication of an experimental prototype by the end of September 2024 and, if the design proved successful, to enter the mass-production phase.

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List of Abbreviations

Abbreviation	Description
μ	Entrainment ratio
ΔP	Pressure drop
η_{mn}	Motive – nozzle efficiency
η_{sn}	Suction – nozzle efficiency
η_{ms}	Mixing – section efficiency
η_d	Diffuser efficiency
s	Isentropic process
A	Cross – sectional area
C	Velocity
e	Evaporator
c	Condenser
tp	Two – phase
Mix	Mixing section
Mt	Motive – nozzle throat
Sh	Superheating
Sc	Subcooling
S	Suction (secondary) nozzle
Opt	Optimum
g	Vapor

Chapter 1: General Considerations

1-1 Introduction

The performance of a vapor-compression refrigeration cycle can be improved through various methods that reduce system energy consumption. One common approach is to use heat exchange to subcool and superheat the working fluid. More recently, researchers have investigated compressor-control strategies based on cooling load as well as the use of an ejector as an expansion device. This section provides a comprehensive review of the ejector as an expansion device in vapor-compression refrigeration cycles.

Vapor-compression refrigeration cycles commonly use capillary tubes and thermostatic expansion valves to reduce refrigerant pressure. Treating this pressure reduction as an isenthalpic process (constant enthalpy) results in a reduction in the cooling capacity of the evaporator because useful energy is lost during throttling. Recovering this energy requires an isentropic process (constant entropy) during expansion. Ejectors can provide conditions closer to this ideal and are therefore used as two-phase ejector expansion devices.

A steam-jet chiller is a type of vapor-based chiller used in air-conditioning applications. The physical operating principle of a steam-jet chiller is fundamentally based on gas dynamics.

In a steam-jet chiller, gas-dynamic principles are used so that gases passing through a specific cycle cool water or another fluid, thereby producing the required refrigeration effect. This type of chiller differs from conventional vapor-compression chillers: compression systems employ compressors and mechanical rotating or reciprocating mechanisms, whereas a steam-jet chiller contains no comparable rotating or reciprocating compression equipment.

1-2 Project Motivation and Significance

A steam chiller is also referred to as a steam-jet refrigeration system. Its operation begins by lowering the boiling point of a fluid through a reduction in the pressure acting on the fluid. In general, lowering the pressure on a fluid reduces its boiling temperature; under sufficiently low pressure, a fluid—particularly water—may approach a phase change and evaporate even near room temperature (about 25 °C). The reverse is also true: increasing the pressure raises the temperature required for evaporation, as can be observed in a household pressure cooker.

This fluid behavior is utilized in a steam chiller. The system contains a horn-shaped ejector into which high-pressure fluid enters from one side. After passing through the nozzle, the fluid enters a low-pressure region, and its velocity increases. This acceleration creates suction in the ejector, drawing in a low-pressure secondary fluid and generating a vacuum that reduces the pressure along the secondary-flow path.

Several methods are available for improving the performance of vapor-compression refrigeration cycles, and such improvements reduce system energy consumption. Heat exchange for refrigerant subcooling and superheating is one common method. More recently, researchers have studied compressor-control and compressor-motor-control strategies based on the cooling load of the conditioned space. The use of an ejector as an expansion device is another alternative for improving cycle performance. This section provides a comprehensive review of the ejector as an expansion device in vapor-compression refrigeration cycles. Conventional vapor-compression systems commonly use capillary tubes, thermostatic expansion valves, and other throttling devices to reduce refrigerant pressure between the condenser and evaporator. Ideally, pressure reduction would occur isentropically, but conventional throttling is approximately isenthalpic (constant enthalpy), which causes a loss of available energy and reduces evaporator cooling capacity. An ejector can recover part of this expansion work by replacing the conventional throttling process with a process that more closely approaches isentropic expansion. Because the working fluid in the diffuser is generally two-phase, this device is commonly referred to as a two-phase ejector; the corresponding system is also called an ejector-expansion refrigeration cycle (EERC).

Ejectors are used in several refrigeration applications. For example, in an ejector refrigeration cycle (ERC), a thermally driven ejector can replace the compressor of a vapor-compression refrigeration cycle. The ERC, introduced by LeBlanc and Parsons in 1901, successfully produced a refrigeration effect using an ejector driven by thermal energy. When steam provides the driving energy, the system is known as a steam-jet refrigeration system. Sokolov and Hershgal proposed an ERC that combined mechanical energy with waste heat; their system could be more economical than a vapor-compression cycle when waste energy is available at sufficiently low cost. The concept of using a two-phase ejector as an expansion device is also not new. In 1931, Gay patented a two-phase ejector intended to reduce throttling losses by replacing a conventional expansion valve with an ejector. In 1966, Kemper et al. proposed modifications involving a pump and heater to raise the pressure and temperature of the liquid stream before it entered the motive nozzle. Newton proposed another variation in 1972 by injecting hot gas from the compressor discharge into the liquid stream before it entered the motive nozzle. A modern numerical analysis of the EERC was later performed by Kornhauser in 1990. Research on ejectors as expansion devices has increased steadily since then. Numerical and experimental studies of EERCs have shown that the coefficient of performance (COP) of vapor-compression refrigeration cycles can be increased. This improvement arises from reduced compressor work and increased evaporator cooling

capacity. Researchers have also investigated the effects of ejector geometric parameters, including motive-nozzle dimensions, mixing-chamber dimensions, constant-area-section dimensions, and diffuser angle. In summary, ejector applications in refrigeration systems can be divided into at least three categories. The first is the use of an ejector as a replacement for the compressor in thermally driven refrigeration systems, an application of growing interest as fossil-fuel resources decline. The second is the use of an ejector in the condenser side of a vapor-compression refrigeration cycle.

The second application is still relatively new. Bergander claimed that his proposed use of an ejector in the condenser line of a vapor-compression refrigeration cycle was the first of its kind. Chen et al. proposed an innovative ejector arrangement in a vapor-compression heat-pump cycle for water heating under low ambient-temperature conditions. In this arrangement, the ejector is positioned in the refrigeration cycle in a manner analogous to locating it between the compressor and condenser. The third application is the use of an ejector to replace a conventional expansion device in order to reduce throttling losses. The present project focuses comprehensively on this third application.

An overview of the three ejector applications in vapor-compression refrigeration cycles is shown below. For the heat-driven refrigeration system in Fig. 1(a), the primary (high-pressure) flow, secondary (low-pressure) flow, and diffuser correspond to state points 1, 6, and 3, respectively. For the ejector located in the condenser line in Fig. 1(b), the primary flow, secondary flow, and diffuser correspond to state points 2, 7, and 4. For the ejector used as an expansion device in Fig. 1(c), these three locations correspond to state points 3, 8, and 5, respectively.

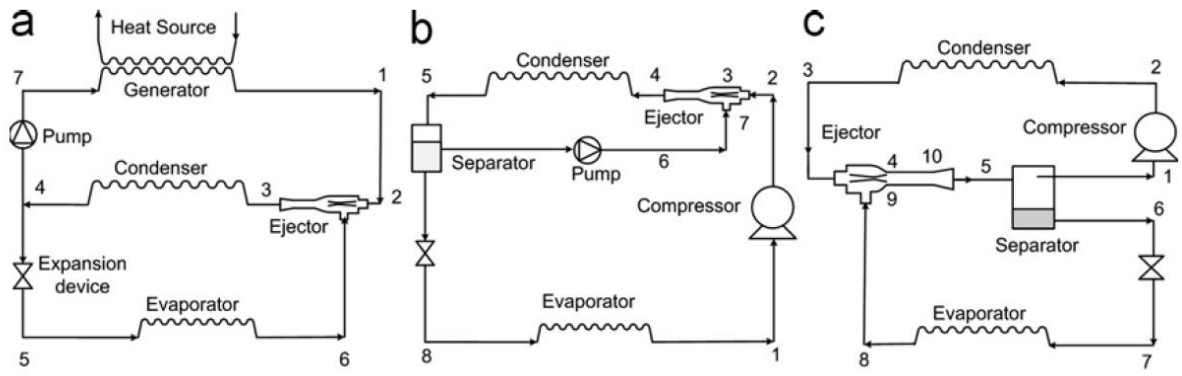


Figure 1: Schematic diagrams of three ejector applications in a vapor-compression refrigeration cycle: (a) ejector in a heat-driven refrigeration system, (b) ejector in the condenser line, and (c) ejector used as an expansion device.

The following figure illustrates the working-fluid phases for the three ejector applications on a P-h (pressure-enthalpy) diagram. It shows that the primary and secondary working-fluid streams are in different thermodynamic phases depending on the application. The operating principle of an ejector is similar to that of a pump. A pump requires external power to raise a fluid to a higher pressure, whereas an ejector uses a high-pressure fluid (the primary flow) to entrain a low-pressure fluid (the secondary flow) and deliver the combined stream at a higher pressure at the diffuser outlet.

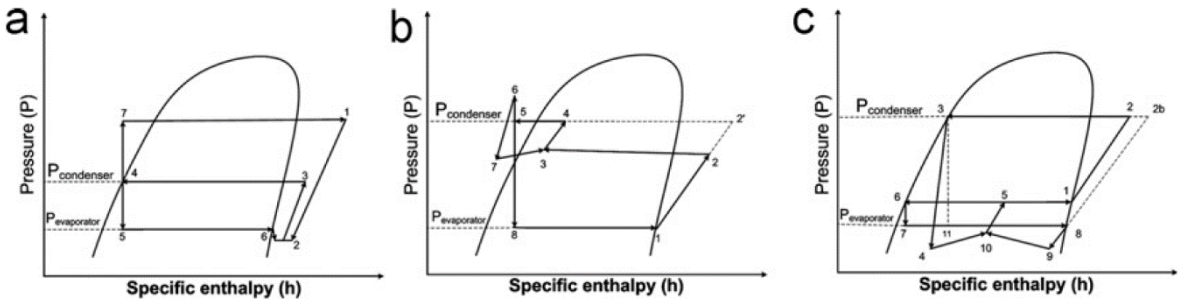


Figure 2: Phase diagrams for three ejector applications in a vapor-compression refrigeration cycle: (a) ejector in a heat-driven refrigeration system, (b) ejector in the condenser line, and (c) ejector used as an expansion device.

The following table provides estimated working-fluid phases and temperature ranges through the ejector at the motive-nozzle outlet, suction inlet, and diffuser outlet. It also summarizes the different applications of ejectors in refrigeration systems.

Table 1: Phase and temperature range of the working fluid in the ejector.

Application on the refrigeration cycle	Nozzle flow	Suction flow	Diffuser flow	Objectives
Heat-driven	**Vapor**	**Vapor**	**Vapor**	To replace the compressor on the vapor compression refrigeration cycle.
Temperature range (°C) (Estimation)				
– Only for air conditioning	> 80	0–15	35–55	
On the condenser	**Vapor**	**Liquid**	**Two-phase**	To improve the COP by reducing the compressor work.
Temperature range (°C) (Estimation)				
– For air conditioning	< 35	< 20	35–55	
– For freezer	< 35	< 20	35–55	
Expansion device	**Liquid**	**Vapor**	**Two-phase**	To improve the COP by lowering the compressor work and increasing the cooling capacity.
Temperature range (°C) (Estimation)				
– For air conditioning	< 5	0–15	> 0, < 15	
– For freezer	< (−5)	(−40)–(−5)	> (−40), < (−5)	

1-3 Operating Principles of an Ejector as an Expansion Device

The following figure shows a constant-area mixing ejector consisting of three main sections: a nozzle, a constant-area mixing section, and a diffuser. Mixing between the high-pressure primary stream and the low-pressure secondary stream occurs within the constant-area section.

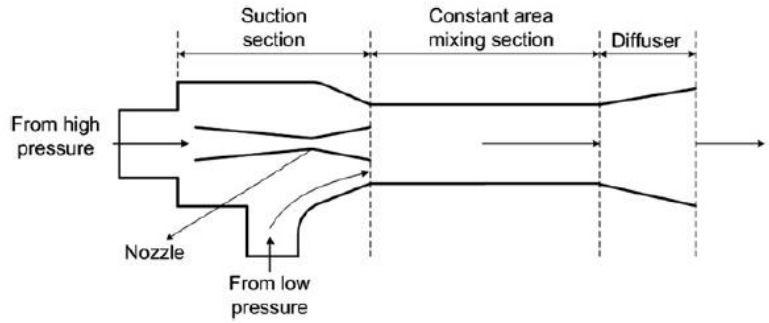


Figure 3: Constant-area mixing ejector

The following figure shows a constant-pressure mixing ejector. It includes a nozzle, a mixing region, a constant-area section, and a diffuser. The primary and secondary streams mix in the mixing section, also referred to as the suction chamber.

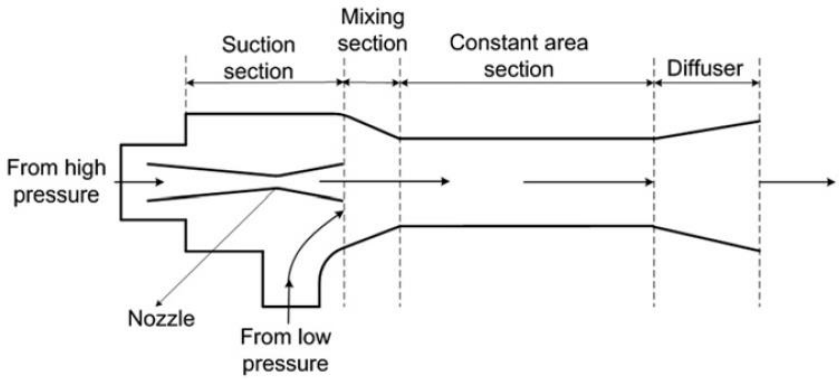


Figure 4: Constant-pressure mixing ejector

The development of mathematical models for internal ejector flow has been reviewed by He et al. Their work describes the evolution and current state of ejector mathematical models. Some studies have reported better performance for constant-pressure mixing ejectors than for constant-area mixing ejectors. Consequently, constant-pressure mixing ejectors have often been used in refrigeration applications, particularly ejector refrigeration systems. However, Yapici and Ersoy found that, at the same operating temperature, a constant-area mixing ejector produced a higher COP than a constant-pressure mixing ejector. Therefore, during the last decade, many numerical and experimental EERC studies have employed constant-area mixing ejectors.

Capillary tubes, thermostatic expansion valves, and automatic expansion valves are expansion devices commonly used in refrigeration systems. Their primary function is to impose a flow restriction that reduces pressure, resulting in a throttling process. Expansion devices can generally be classified as fixed-restriction or variable-restriction devices. A capillary tube is a fixed-restriction expansion device with a small diameter. Thermostatic and automatic expansion valves are variable-restriction devices whose opening varies according to the evaporator cooling load. The capillary tube is the simplest and least expensive expansion device and is commonly used in low-capacity refrigeration systems. For

medium- to high-capacity systems, thermostatic or automatic expansion valves are generally used. Capillary tubes and other conventional expansion devices dissipate the kinetic energy associated with the pressure reduction. The throttling process is theoretically approximated as isenthalpic. Using an ejector as the expansion device allows part of the expansion process to approach an isentropic path, as shown in the following figure. The isenthalpic process extends from state point 3 to 11, whereas the idealized isentropic process extends from state point 3 to 4.

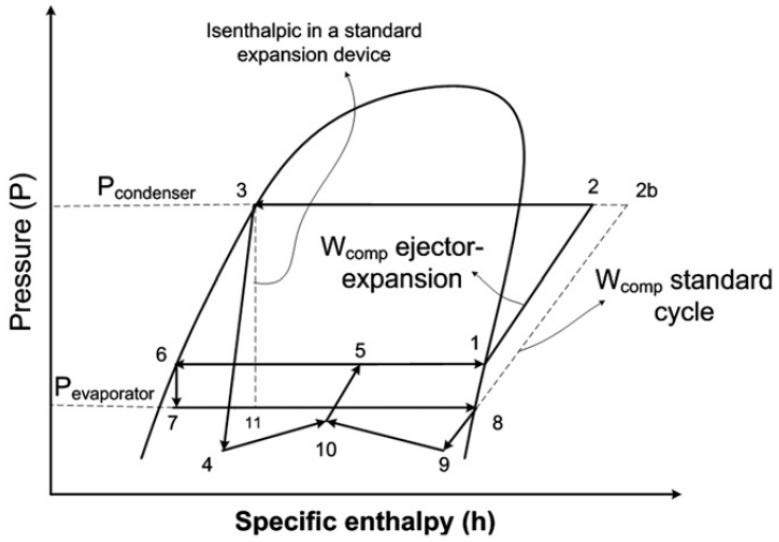


Figure 5: Standard P-h diagram for the EERC and the conventional cycle

The figure above presents the EERC and the conventional refrigeration cycle on a P-h diagram. In the conventional cycle, the refrigerant follows state points 8, 2b, 3, 11, and 8. In the EERC, two flow paths are present: the primary flow and the secondary flow. The primary flow circulates through the compressor, heat exchanger, ejector, and separator (state points 1, 2, 3, 4, 10, 5, and 1), while the secondary flow circulates through the expansion valve, heat exchanger, ejector, and separator (state points 6, 7, 8, 9, 10, 5, and 6). The primary and secondary flows mix in the constant-area region and diffuser (between state points 10 and 5). As shown in the figure, the pressure at state point 1 is higher than the compressor suction pressure of the conventional cycle at state point 8. This reduces compressor work in the ejector-expansion cycle relative to the conventional cycle.

The operating principle of a two-phase ejector used as an expansion device is similar to that of ejectors in other applications: a high-pressure primary stream entrains a low-pressure secondary stream in the suction section and raises it to a higher pressure in the diffuser. The following figure shows the refrigerant flow as well as pressure and velocity characteristics inside the ejector. The motive, or primary, flow accelerates and expands through the motive nozzle from state point 3 to 4. Its very high velocity produces a pressure reduction at state point 4, which draws the secondary fluid from state point 8 to 9. The two streams mix and become a single stream in the constant-area mixing section. The mixed flow then moves from state point 10 into the diffuser. In the diffuser, the refrigerant velocity decreases as the pressure rises toward state point 5.

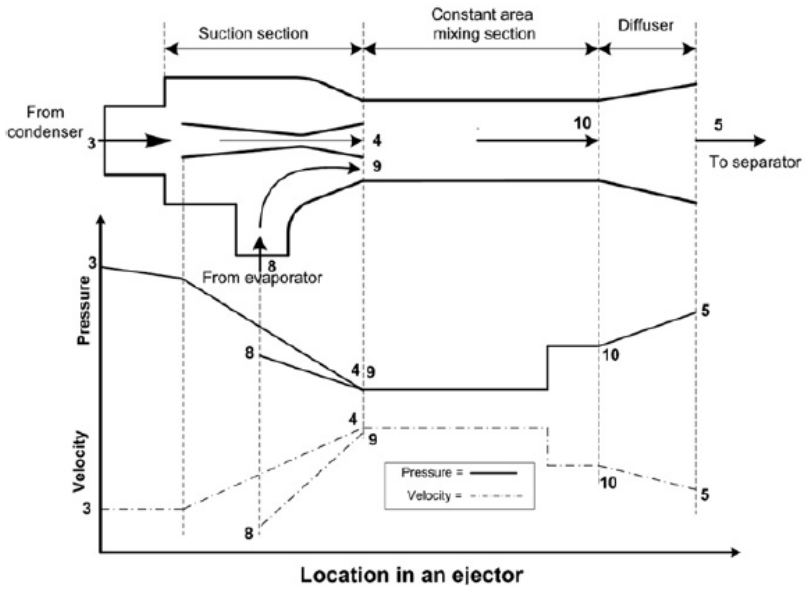


Figure 6: Refrigerant flow, pressure, and velocity characteristics inside the ejector

The following schematic shows the conventional cycle and an EERC. Refrigerant at state point 5 enters the separator. The vapor refrigerant leaving the separator circulates through the compressor (from state point 1 to 2) as the primary flow, while the liquid refrigerant from the separator circulates through the expansion valve and evaporator (state points 6, 7, and 8) as the secondary flow.

1-4 Refrigerant Used in This Project

The refrigerant used for the thermodynamic analysis of this cycle is R1234yf. R1234yf (chemical name: 2,3,3,3-tetrafluoropropene) is a new-generation refrigerant that, because of its environmental advantages and high performance, is used as an alternative to R134a in air-conditioning systems and other refrigeration applications. The two refrigerants have broadly similar cooling capacities and energy consumption, with only minor differences.$C_3H_2F_4$

R1234yf belongs to the hydrofluoroolefin (HFO) family. It has a very low global warming potential (GWP), reported here as below 4.4, which makes it an environmentally favorable option. By comparison, R134a has a much higher GWP, approximately 1370. Regarding ozone-depletion potential, the ODP of both R134a and R1234yf is zero; therefore, neither refrigerant directly contributes to depletion of the ozone layer. This characteristic is compatible with international environmental regulations such as the Montreal Protocol.

R1234yf was also selected so that the thermodynamic-cycle simulation could be validated against results reported in available peer-reviewed studies. By comparing the present simulation results with those references, the accuracy of the analysis and the resulting predictions could be assessed.

1-5 Review of Previous Research

Because of the adverse environmental effects of many synthetic refrigerants, CO_2 has been considered an environmentally preferable working fluid. Expansion losses in CO_2 vapor-compression refrigeration cycles are also relatively high compared with cycles using conventional refrigerants. For example, in conventional vapor-compression systems the expansion pressure drop may typically be from about 2 to 0.2 MPa, depending on the working fluid and evaporation temperature. In transcritical CO_2 refrigeration cycles, by contrast, expansion may occur from roughly 10 to 2 MPa. Therefore, in principle, the potential for recovering expansion losses generated by the throttling process is greater in transcritical CO_2 systems than in conventional cycles.

Liu et al. presented a numerical analysis of an ejector-expansion refrigeration system using CO_2 as the refrigerant and applying the Kornhauser iteration method. The ejector was used in place of an expansion valve in the transcritical cycle to recover part of the kinetic-energy loss associated with throttling. They found that compressor suction pressure was higher than in the conventional cycle, thereby reducing the compression ratio and compressor work in the ejector-expansion cycle. COP improvement in the transcritical cycle ranged from 6% to 14% relative to the conventional cycle. Ozaki et al. experimentally investigated an automotive air-conditioning system using R744 and reported a 20% COP improvement relative to the conventional cycle. They also showed that a two-phase ejector provides an advantage in controlling the high-side pressure of a transcritical system.

Li and Groll performed a thermodynamic numerical analysis using a constant-pressure ejector model. The effects of entrainment ratio and pressure drop in the suction chamber were investigated. Under the specified conditions, the ejector-expansion cycle produced a COP improvement of approximately 7-18% relative to the conventional cycle. A reduction in entrainment ratio increased the suction pressure, which reduced the compression ratio and therefore increased the COP improvement. They also found that the COP improvement increased as the pressure drop in the suction chamber increased. The suction-chamber pressure drop is governed by ejector geometry and operating temperatures. For a transcritical CO_2 cycle under typical air-conditioning operating conditions, the system improvement over the conventional cycle could exceed 16%.

Deng et al. conducted energy and exergy analyses of a transcritical ejector-expansion cycle. Their energy analysis indicated a 22% COP improvement relative to the conventional cycle. Exergy losses of the individual components in the ejector-expansion and conventional cycles at a high-side pressure of 8.7 MPa are shown in the following figure.

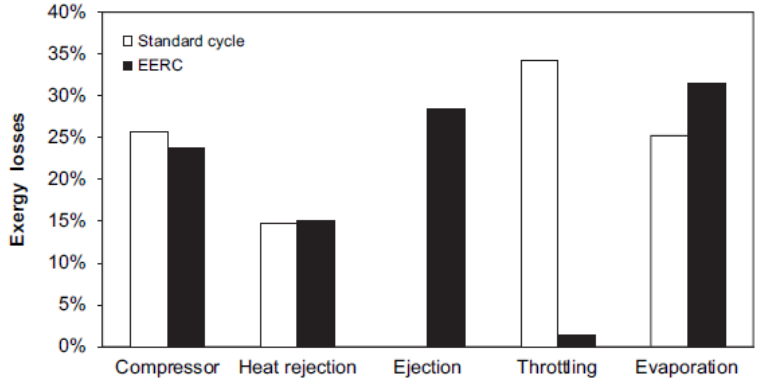


Figure 7: Exergy losses in the conventional cycle and the EERC at a high-side pressure of approximately 8.7 MPa.

The figure shows that throttling-process exergy destruction in the conventional cycle accounts for 34.29% of the total exergy destruction. In the ejector-expansion cycle, the corresponding expansion-related exergy losses are reported as 1.36% and 28.38%, giving a combined value of 29.7% of the total system exergy loss, lower than in the conventional cycle. Compressor exergy destruction also decreases from 25.69% of total exergy destruction in the conventional cycle to 23.73% in the ejector-expansion cycle.

Yari and Sirousazar theoretically analyzed a two-stage transcritical CO_2 ejector-expansion refrigeration cycle. They proposed a new system incorporating an internal heat exchanger and an intercooler. Their simulations showed that the COP and second-law efficiency of the proposed cycle increased by approximately 55.5% and 26%, respectively, at an evaporation temperature of 10 °C and a gas-cooler outlet temperature of 40 °C. They also found that the total exergy destruction of the new two-stage ejector-expansion cycle was 4% lower than that of the conventional cycle.

Elbel and Hrnjak were among the first researchers to introduce a variable-geometry two-phase ejector by installing a needle in the motive nozzle to adjust the nozzle-throat area. Their results showed that both cooling capacity and COP could be maximized by adjusting the needle opening at the nozzle throat. Improvements in cooling capacity and COP reached approximately 8% and 7%, respectively. The highest COP was obtained when the nozzle-throat area was reduced. This result was similar to experimental findings reported by Chaiwongsa and Wongwises, who found that an ejector-expansion cycle performed best with a nozzle-throat diameter of 0.8 mm compared with 0.9 and 1.0 mm. In addition to introducing a variable-area nozzle, Elbel and Hrnjak investigated the effect of diffuser angle on ejector-expansion-cycle performance. Three different diffuser angles—5°, 10°, and 15°—were tested, and the smallest angle, 5°, produced the best ejector performance. Elbel also reviewed ejector applications in refrigeration and reported experimental results for a CO_2 ejector-expansion refrigeration system. Using diffuser modules similar to those in the earlier study, four different constant-area mixing-section lengths of 7.5, 33.5, 57.5, and 82.5 mm were tested. The constant-area length had a significant effect on ejector performance: the shortest section produced an efficiency of up to 15%, whereas efficiency decreased to about 7% for the longest section. Secondary flow and pressure recovery increased for the shorter constant-area section. Improvements in cooling capacity and COP relative to the conventional cycle were approximately 7% and 8%, respectively, consistent with earlier findings.

The following table summarizes studies of ejector-expansion refrigeration cycles using CO_2 as the refrigerant.

Table 2:

Authors	Working fluid	COP improvement	Remarks
Liu et al. (2002) [49]	R744 (CO_2)	6–14%	Numerical.
Ozaki et al. (2004) [57]	R744 (CO_2)	20%	Experimental on the automotive air conditioning.
Li and Groll (2005) [50]	R744 (CO_2)	16%	Numerical on the air conditioning system.
Deng et al. (2007) [58]	R744 (CO_2)	22%	Numerical.
Yari & Sirousazar (2008) [59]	R744 (CO_2)	55.5%	Numerical analysis on two-stage refrigeration cycle using internal heat exchanger and an intercooler.
Elbel & Hrnjak (2008) [11]	R744 (CO_2)	7%	Experimental on the air conditioning system. The best result is for 5 °C angle of diffuser.
Elbel (2011) [12]	R744 (CO_2)	7%	Experimental: the diffuser angle and the mixing section length have significant effect on the COP improvement.

Table 2. Summary of studies on using an ejector as an expansion device in vapor-compression refrigeration cycles with carbon dioxide (CO_2) as the refrigerant

Chapter 2: Project Methodology

2-1 Introduction

Steam-jet refrigeration systems, like other refrigeration systems, include major components such as an evaporator, condenser, and compression device. Their fundamental difference from conventional refrigeration systems is the use of a steam jet instead of a mechanical compressor to generate suction and compression. In such systems, water is used as the refrigerant, and refrigeration is produced by evaporating part of the water in the evaporator.

The steam jet plays the principal role in producing compression and transferring the vapor. Because steam jets can handle large volumetric flow rates effectively, this type of system is particularly suitable for large refrigerant (water) volume flows. The following figure presents a general schematic of a steam-jet refrigeration system.

Because a steam-jet refrigeration system does not use a conventional compressor, it can be economically attractive for capacities above approximately 100 refrigeration tons. This characteristic makes such systems suitable for high-capacity applications. Other applications include concentration of materials at low temperature while preserving product quality and dehumidification of fruits and vegetables.

By applying thermodynamic principles, steam-jet refrigeration systems can provide effective performance under a range of operating conditions. Eliminating a mechanical compressor can reduce initial and maintenance costs, and lower electrical energy consumption can reduce operating expenses. For these reasons, steam-jet refrigeration systems have been considered an efficient technology in refrigeration applications.

2-2 Thermodynamic Equations Governing the Cycle

In this section, the thermodynamic equations governing the cycle are examined. Reference [1] is used to derive the governing equations for each component. As noted in the introduction, this cycle has two major differences from other ejector-based vapor-compression refrigeration cycles:

1. No pump is used in this refrigeration cycle to increase the COP.
2. The ejector used in this cycle is a two-phase ejector, whereas most ejectors constructed to date have been single-phase ejectors whose working fluid remains in the vapor phase.

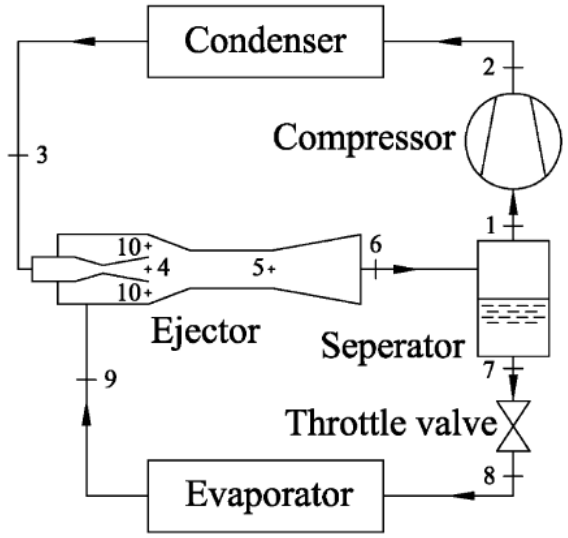


Figure 9: Overall schematic of the ejector-based vapor-compression refrigeration cycle without a pump or heat recuperator

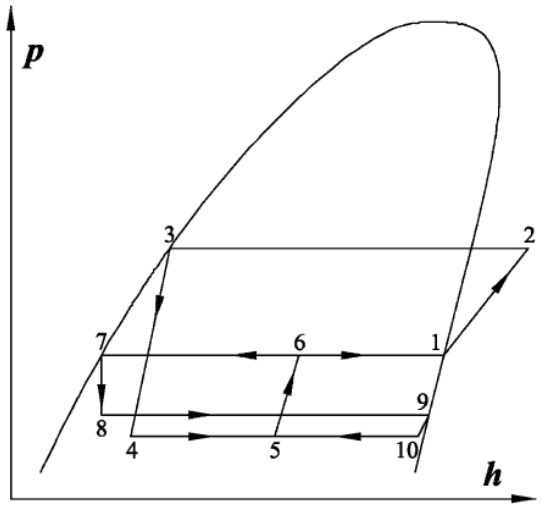


Figure 8: Predicted P-h process diagram based on Ref. [1]

Assumptions used in the thermodynamic analysis of the cycle [1]:

1. Steady-state operating conditions are assumed for the system.
2. Heat transfer with the surroundings is assumed to occur only in the condenser.
3. Pressure drops in the connecting pipes are neglected.
4. The fluids leaving the condenser and evaporator are assumed to be saturated.
5. The vapor and liquid streams leaving the separator are assumed to be saturated.
6. The compressor has a specified isentropic efficiency, defined as an input parameter.
7. The process through the throttling valve is treated according to the stated expansion assumption.
8. Flow through the ejector is assumed to be one-dimensional and uniform.
9. Kinetic-energy effects at the ejector inlet and outlet are neglected.
10. The motive and suction streams reach the same pressure at the entrance to the ejector constant-area mixing section, and no mixing occurs between the two streams before entering the mixing section.
11. Frictional effects and other irreversibilities are represented through efficiencies assigned to the nozzle, mixer, and diffuser.

Using the eleven assumptions listed above, the thermodynamic equations governing each component of the cycle are now examined step by step.

Governing Equations for the Condenser:

At the condenser outlet, the refrigerant is saturated liquid; therefore, its vapor quality is zero.

$$\begin{cases} T_c = T_{Condesncing\ Process} \\ P_c = P_{Condesncing\ Process} \end{cases}$$

$$P_3 = P_c = P(T_c, x = 0) \quad (1)$$

$$h_3 = h(T_c, x = 0) \quad (2)$$

$$S_3 = S(T_c, x = 0) \quad (3)$$

Governing Equations for the Evaporator:

At the evaporator outlet, the refrigerant is saturated vapor. Therefore:

$$\begin{cases} T_e = T_{Evaporation\ Process} \\ P_e = P_{Evaporation\ \ Process} \end{cases}$$

$$P_9 = P_e = P(T_e, x = 1) \quad (4)$$

$$h_9 = h(T_e, x = 1) \quad (5)$$

$$S_9 = S(T_e, x = 1) \quad (6)$$

The ejector is a key component and can be classified into two main types: constant-pressure mixing and constant-area mixing ejectors. Many studies have reported better performance for constant-pressure mixing ejectors than for constant-area ejectors [2,3]. Therefore, a constant-pressure mixing ejector is used in this study.

Governing Equations for the Ejector:

1- Motive-Nozzle Outlet

$$P_4 = P_9 - \Delta P \quad (7)$$

$$h_{4,s} = h(P_4, S_3) \quad (8)$$

$$h_4 = h_3 - \eta_{mn}(h_3 - h_{4,s}) \quad (9)$$

$$c_4 = \sqrt{2(h_3 - h_4)} \quad (10)$$

$$A_4 = \frac{1}{(1+\mu)\rho_4 c_4} \quad (11)$$

2- Suction-Nozzle Outlet

$$P_{10} = P_9 - \Delta P \quad (12)$$

$$h_{10,s} = h(P_{10}, S_9) \quad (13)$$

$$h_{10} = h_9 - \eta_{sn}(h_9 - h_{10,s}) \quad (14)$$

$$c_{10} = \sqrt{2(h_9 - h_{10,s})} \quad (15)$$

$$A_{10} = \frac{\mu}{(1+\mu)\rho_{10}c_{10}} \quad (16)$$

3- Mixing Section

$$P_5 = P_4 = P_{10} \quad (17)$$

$$c_5 = \sqrt{\eta_{ms}}\left(\frac{1}{1+\mu}c_4 + \frac{\mu}{1+\mu}c_{10}\right) \quad (18)$$

$$h_5 = \frac{1}{1+\mu}\left(h_4 + \frac{c_4^2}{2}\right) + \frac{\mu}{1+\mu}\left(h_{10} + \frac{c_{10}^2}{2}\right) - \frac{c_5^2}{2} \quad (19)$$

$$S_5 = S(P_5, h_5) \quad (20)$$

4- Diffuser Outlet

$$h_6 = h_5 + \frac{c_5^2}{2} \quad (21)$$

$$h_{6,s} = h_5 + \eta_d(h_6 - h_5) \quad (22)$$

$$P_6 = P(h_{6,s}, S_5) \quad (23)$$

Note: The following equality must ultimately be satisfied:

$$x_6 = \frac{1}{1+\mu} \quad (24)$$

$$AR = \frac{A_4 + A_{10}}{A_4} \quad (25)$$

$$PLR = \frac{P_6}{P_9} \quad (26)$$

Once ejector performance has been determined, the performance of the remaining components is calculated using conventional thermodynamic relations.

Governing Equations for the Compressor [1]

$$h_{2,s} = h(P_2, S_1) \quad (27)$$

$$h_2 = h_1 + \frac{h_{2,s} - h_1}{\eta_{comp}} \quad (28)$$

$$\eta_{comp} = 0.874 - 0.0135\frac{P_c}{P_e} \quad (29)$$

The compressor isentropic efficiency is determined using the empirical correlation proposed by Brunin et al. [3].

Governing Equations for the Throttling Valve

$$h_8 = h_7 \tag{30}$$

The compressor work and the evaporator heat-transfer rate can also be obtained from the following relations:

$$W_{comp} = \frac{h_2 - h_1}{1 + \mu} \tag{31}$$

$$Q_{evap} = \frac{\mu(h_9 - h_8)}{1 + \mu} \tag{32}$$

Cycle Coefficient of Performance

$$COP = \frac{Q_{evap}}{W_{comp}} = \frac{\mu(h_9 - h_8)}{h_2 - h_1} \tag{33}$$

Cycle Refrigeration Capacity

$$VCC = \frac{\mu(h_9 - h_8)}{\nu_1} \tag{34}$$

The improvements in COP and VCC of the EERC relative to a conventional refrigeration cycle can then be defined as follows:

$$COP_{imp} = \frac{COP - COP_s}{COP_s} \tag{35}$$

$$VCC_{imp} = \frac{VCC - VCC_s}{VCC_s} \tag{36}$$

Here, the corresponding quantities denote the COP and volumetric cooling capacity of a conventional vapor-compression refrigeration cycle operating at the same evaporation and condensation temperatures. $COP_s VCC_s$

*Table*3 *: Governing thermodynamic equations of the cycle together with the initial assumptions for each component*

Component	Governing Thermal Equations		Assumptions
Condensor	$P_3 = P_c = P(T_c, x = 0)$ $h_3 = h(T_c, x = 0)$ $S_3 = S(T_c, x = 0)$		1. At the condenser outlet, the refrigerant is saturated liquid; therefore, its vapor quality is zero.
Evaporator	$P_9 = P_e = P(T_e, x = 1)$ $h_9 = h(T_e, x = 1)$ $S_9 = S(T_e, x = 1)$		1. Pressure variation through the evaporator is neglected.

Ejector	Motive Nozzle outlet	$P_4 = P_9 - \Delta P$ $h_{4,s} = h(P_4, S_3)$ $h_4 = h_3 - \eta_{mn}(h_3 - h_{4,s})$ $c_4 = \sqrt{2(h_3 - h_4)}$ $A_4 = \frac{1}{(1+\mu)\rho_4 c_4}$		Complete mixing between the low-velocity and high-velocity streams is assumed after they leave the nozzles.
	Suction Nozzle Outlet	$P_{10} = P_9 - \Delta P$ $h_{10,s} = h(P_{10}, S_9)$ $h_{10} = h_9 - \eta_{sn}(h_9 - h_{10,s})$ $c_{10} = \sqrt{2(h_9 - h_{10,s})}$ $A_{10} = \frac{\mu}{(1+\mu)\rho_{10} c_{10}}$		
	Mixing section	$P_5 = P_4 = P_{10}$ $c_5 = \sqrt{\eta_{ms}}\left(\frac{1}{1+\mu} c_4 + \frac{\mu}{1+\mu} c_{10}\right)$ $h_5 = \frac{1}{1+\mu}\left(h_4 + \frac{c_4^2}{2}\right) + \frac{\mu}{1+\mu}\left(h_{10} + \frac{c_{10}^2}{2}\right) - \frac{c_5^2}{2}$ $S_5 = S(P_5, h_5)$		
	Diffuser Outlet	$h_6 = h_5 + \frac{c_5^2}{2}$ $h_{6,s} = h_5 + \eta_d(h_6 - h_5)$ $P_6 = P(h_{6,s}, S_5)$		
Convergence Condition		$x_6 = \frac{1}{1+\mu}$		In this section, the value obtained at the diffuser outlet is compared with the corresponding calculated value so that the required convergence quantity can be evaluated.$x_6\mu$
Compressor		$h_{2,s} = h(P_2, S_1)$ $h_2 = h_1 + \frac{h_{2,s} - h_1}{\eta_{comp}}$		

	$\eta_{comp} = 0.874 - 0.0135\frac{P_c}{P_e}$		
Thottle valve	$h_8 = h_7$		
gas–liquid separator outlets	$h_1 = h(P_6, x = 1)$ $h_7 = h(P_6, x = 0)$		
Output Result	$W_{comp} = \frac{h_2 - h_1}{1 + \mu}$ $Q_{evap} = \frac{\mu(h_9 - h_8)}{1 + \mu}$ $COP = \frac{Q_{evap}}{W_{comp}} = \frac{w(h_9 - h_8)}{h_2 - h_1}$ $VCC = \frac{\mu(h_9 - h_8)}{v_1}$		
Comparison between classical compression refrigeration cycle and ejector compression refrigeration cycle	$COP_{imp} = \frac{COP - COP_s}{COP_s}$ $VCC_{imp} = \frac{VCC - VCC_s}{VCC_s}$		

After deriving the thermodynamic equations governing the cycle in the previous sections, REFPROP is used to obtain the fluid properties at the different state points of the cycle.

A question may arise as to why EES, one of the most commonly used programs for solving thermodynamic-cycle equations, was not used. The reason is the iterative nature of the present problem. To determine a property such as the refrigerant quality at the diffuser outlet, repeated iterations are required until convergence is achieved. Implementing this type of nested iterative algorithm is comparatively less convenient in EES for the present workflow.$x_6 for - While$

2-2-1 Computational Algorithm

Following the method proposed in Ref. [1], the main inputs to the problem are the condenser temperature, evaporator temperature, and the specified pressure difference between state points 9 and 10, i.e., between the suction-nozzle inlet and the evaporator outlet. This pressure difference must be prescribed because it is directly related to ejector design, as discussed later. In addition, four ejector efficiencies are required as input parameters: diffuser efficiency, mixing-section efficiency, suction-nozzle efficiency, and motive-nozzle efficiency.

The entrainment ratio is also defined in terms of the mass-flow rates of the primary and secondary branches. The primary and secondary cycles must therefore be clearly identified.μ

In the following figure, the red-shaded region represents the primary branch of the cycle with its corresponding mass flow rate, while the blue-shaded region represents the secondary branch with its own mass flow rate. The entrainment ratio is then defined as follows:$\dot{m}_p \dot{m}_s \mu$

$$\mu = \frac{\dot{m}_s}{\dot{m}_p} \tag{37}$$

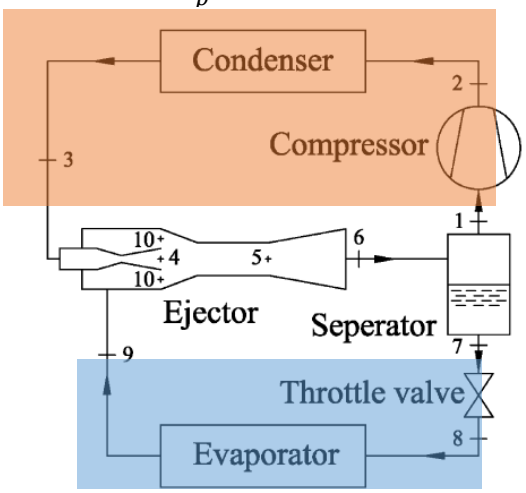


Figure 10: Main cycle with the defined primary and secondary mass flow rates

Because the ejector outlet mass flow rate at state point 6 is the sum of the primary and secondary mass flow rates, the following relation applies:

$$\dot{m}_6 = \dot{m}_s + \dot{m}_p \tag{38}$$

As explained previously and shown in the figure, the fluid circulating in the primary branch corresponds to the vapor portion of the saturated mixture, whereas the fluid in the secondary branch corresponds to the liquid portion of the saturated mixture.

Therefore, as indicated in the corresponding schematic, the fluid entering the primary cycle is saturated vapor and the fluid entering the secondary cycle is saturated liquid. Accordingly, the following relations can be written:

$$x_6 = \frac{m_{vapor}}{m_{total}} = \frac{\frac{m_{vapor}}{dt}}{\frac{m_{total}}{dt}} = \frac{\dot{m}_{vapor}}{\dot{m}_{total}} = \frac{\dot{m}_{vapor}}{\dot{m}_{Liquid} + \dot{m}_{vapor}} \tag{39}$$

$$\begin{cases} \dot{m}_{vapor} = \dot{m}_p \\ \dot{m}_{liquid} = \dot{m}_s \end{cases} \rightarrow \quad x_6 = \frac{\dot{m}_p}{\dot{m}_s + \dot{m}_p} = \frac{1}{\frac{\dot{m}_s}{\dot{m}_p} + 1}$$

$$\mu = \frac{\dot{m}_s}{\dot{m}_p} \rightarrow \quad x_6 = \frac{1}{1 + \mu} \tag{40}$$

To calculate the fluid properties before the state at point 6 and before the ratio of the primary and secondary mass flow rates is known, an initial estimate of the mass-flow-rate ratio is required to evaluate the properties at state point 5, which is immediately upstream of the ejector outlet. This ratio is introduced through the following relation:

State Point 5: Mixing region inside the ejector.

$$P_5 = P_4 = P_{10} \tag{41}$$

$$c_5 = \sqrt{\eta_{ms}}\left(\frac{1}{1+\mu}c_4 + \frac{\mu}{1+\mu}c_{10}\right) \tag{42}$$

$$h_5 = \frac{1}{1+\mu}\left(h_4 + \frac{c_4^2}{2}\right) + \frac{\mu}{1+\mu}\left(h_{10} + \frac{c_{10}^2}{2}\right) - \frac{c_5^2}{2} \tag{43}$$

$$S_5 = S(P_5, h_5) \tag{44}$$

As can be seen from the equations, the desired properties cannot be obtained unless the mass-flow-rate ratio is known. Therefore, at the beginning of the cycle analysis, an initial value of this ratio is assumed. After the properties at state point 6 have been calculated, the ratio can be recalculated from the resulting state.$c_5 S_5 \mu\mu x_6$

$$h_6 = h_5 + \frac{c_5^2}{2} \tag{45}$$

$$h_{6,s} = h_5 + \eta_d(h_6 - h_5) \tag{46}$$

$$x_6 = x(h_6, P_6) \tag{47}$$

After obtaining the updated value using Eq. (47), the following relation is obtained:x_6

$$x_6 = \frac{1}{1+\mu} \tag{48}$$

For the initially assumed value to be valid, this relation must be satisfied. Accordingly, after the fluid properties at state point 6 are obtained, this equation is used as the convergence condition.μ

In this project, the convergence condition is applied by calculating an updated value of the relevant mass-flow ratio from the governing relation and comparing it with the initially assumed value. If the difference between the two values is smaller than the prescribed error tolerance—set to 0.0001 in this project—the calculated value is accepted and the initial assumption is considered converged. Otherwise, the assumed value is adjusted iteratively until the following convergence criterion is satisfied:$\frac{1}{1+\mu} x_6 x'_6 x_6 x_6 x'_6 x_6 \mu' \mu$

Convergence Criterion for the Project:

$$if: |\mu - \mu'| < error \quad \rightarrow \quad \mu\ has\ Correct\ value!$$

$$else: \mu\ Should\ be\ change!$$

This iterative convergence check must therefore be performed for every set of cycle input conditions in order to obtain the correct system solution.

2-2-2 Determination of the Initial Conditions

Based on the procedure described in the previous section, an initial estimate of the mass-flow-rate ratio is required at the start of each cycle calculation. The initial value recommended in Ref. [1] is used to accelerate convergence. Together with the remaining prescribed parameters, this forms the set of initial conditions for the problem.$\mu\mu = 0.774$

$$\begin{cases} T_c = 30°C \\ T_e = 5°C \\ \eta_{mn} = 0.9 \\ \eta_{sn} = 0.85 \\ \eta_d = 0.85 \\ \eta_{ms} = 0.85 \\ \Delta P = 0.5\ kPa \\ \mu = 0.774 \end{cases}$$

The selected temperatures and ejector-component efficiencies are based on ejector fabrication and operating requirements. In particular, the ejector must be capable of operating at the specified evaporator and condenser temperatures used in the design.

2-2-3 Procedure for Applying the Convergence Condition

The convergence procedure used in this project begins by selecting an initial step size of 10. If the initial estimate is not correct, the following cases are considered:

Case 1: The initial guess is greater than the value obtained from the governing equations.

In this case, the assumed value is reduced by the current step size.

Case 2: The initial guess is smaller than the value obtained from the governing equations.

In this case, the assumed value is increased by the current step size.

The step size used in the solution is then divided by 10.

The above procedure is repeated until the difference between the assumed value and the value obtained from the equations becomes smaller than the specified error tolerance.

The resulting convergence flowchart is shown below:

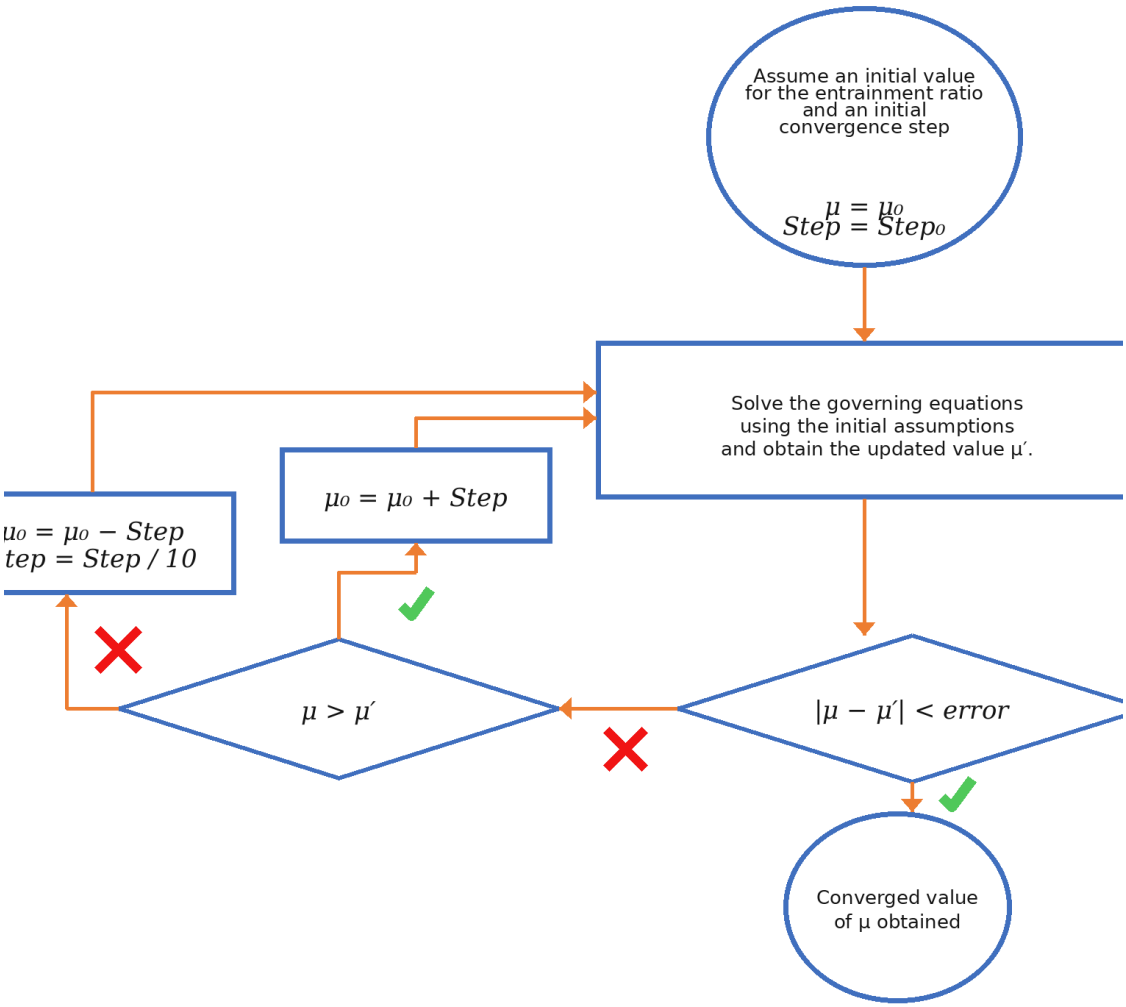


Figure 11: Convergence flowchart for the ejector-based vapor-compression refrigeration cycle without a heat recuperator

2-3 Investigation of the Effect of Adding a Heat Recuperator

A heat recuperator (heat exchanger) is used to recover heat from one fluid stream and transfer it to another stream within the cycle. This can improve cycle performance. Because the objective of this project is to increase the COP as much as possible, the effect of adding a heat recuperator is investigated. An effectiveness of 80% is assumed for the heat exchanger.

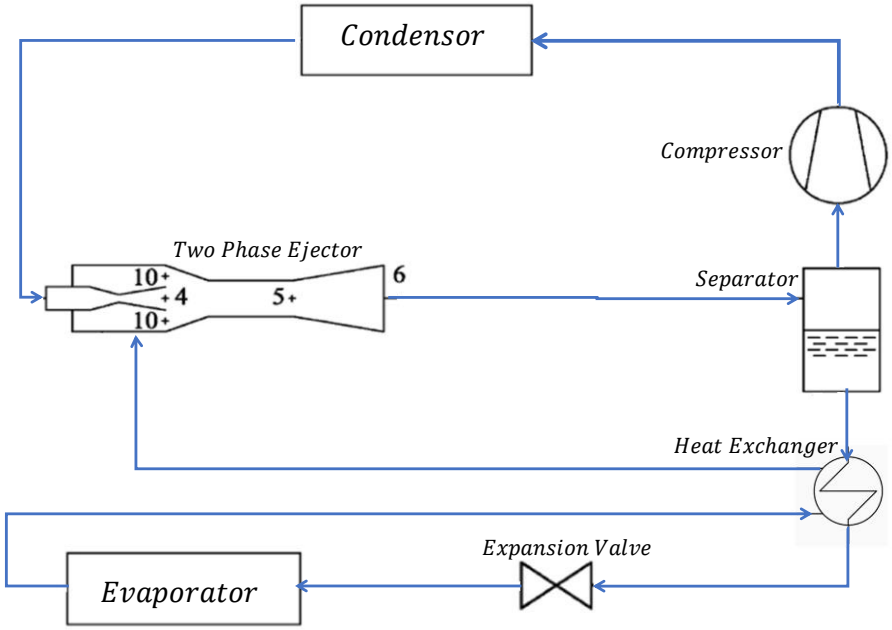


Figure 12: Vapor-compression refrigeration cycle including a two-phase ejector and heat recuperator

2-3-1 Effect of the Heat Recuperator on Cycle Performance

12. Increase in the Temperature of the Fluid Entering the Ejector:

    Using the heat recuperator increases the temperature of the fluid entering the ejector at the corresponding state point. According to the cycle model used here, this additional thermal energy can influence the low-pressure stream and the ejector operating condition.

13. Reduction in Compressor Work Requirement:

    When heat is transferred between the hotter and colder streams in the heat recuperator, the thermodynamic state of the fluid entering the compressor is altered. In the present cycle model, this is expected to reduce the compressor work required for compression.

14. Improvement of Heat Transfer in the Evaporator:

    The fluid leaving the heat recuperator enters the subsequent expansion and evaporation process at a modified thermodynamic state. This can improve the heat-transfer process in the evaporator and, consequently, the overall cycle performance.

15. Reuse of Recovered Heat:

    The heat recuperator transfers recovered heat internally between the two fluid streams, allowing energy that would otherwise remain unused within one stream to be reused within the cycle and potentially improving system performance.

2-3-2 Thermodynamic Equations Governing the Cycle after Adding the Heat Recuperator

The thermodynamic equations of the cycle in this configuration are largely similar to those of the cycle without a heat recuperator. The principal difference is the addition of the equations governing the heat exchanger itself. Therefore, the following relations are added to the previously presented set of cycle equations.

Governing Thermodynamic Equations for the Heat Exchanger:

$$Energy\ Balance\ Equation:\ \dot{m}_7(h_7 - h_7') = \dot{m}_9(h_9' - h_9) \quad (49)$$

$$From\ mass\ Balance: \dot{m}_7 = \dot{m}_9 \quad (50)$$

$$so\ we\ have: (h_7 - h_7') = (h_9' - h_9) \quad (51)$$

$$Efficiency\ Equation: \eta = \frac{T_7 - T_7'}{T_7 - T_9} \quad (52)$$

2-3-3 Computational Algorithm for Simulating the Cycle in REFPROP

The thermodynamic-cycle algorithm implemented in this section is very similar to the algorithm described previously for the cycle without a heat recuperator.

The same basic initial assumptions are retained; however, one important additional requirement is an initial guess for the temperature at state point 7. The reason for introducing this guessed temperature is explained below.$T_{G\,7}$

Initial Conditions Required for Cycle Simulation:

$$\begin{cases} T_c = 30°C \\ T_e = 5°C \\ \eta_{mn} = 0.90 \\ \eta_{sn} = 0.85 \\ \eta_d = 0.85 \\ \eta_{ms} = 0.85 \\ \eta_{HE} = 0.80 \\ \Delta P = 0.5\ kPa \\ \mu = 0.774 \\ T_{G\,7} = 5°C \end{cases}$$

2-3-3-1 Analysis of Cycle Components and Required Assumptions

Condenser and Evaporator

Using the prescribed initial conditions together with the eleven assumptions introduced at the beginning of the project, the properties at state points 3 and 9 can be determined directly and remain fixed for a given simulation.

The heat exchanger is analyzed next. The pressure is assumed to remain constant through each side of the heat exchanger; therefore:

$$P_9 = P_9' \quad (53)$$

To continue the solution, one independent property at the required state point must initially be assumed so that all fluid properties at that point can be evaluated. The assumed value is then recalculated using the heat-exchanger equations. If the difference between the assumed and recalculated values is sufficiently small, the cycle solution is considered converged with respect to this condition.$9'$

In this section, the temperature at the specified state point is used as the guessed property. With pressure and temperature known at that point, the cycle analysis can proceed.$T_9'9'$

Ejector

Using the properties at state points 9 and 3, the properties at state point 4, corresponding to the primary-nozzle outlet, can be determined:

$$P_4 = P_9' - \Delta P \tag{54}$$

$$h_{4,s} = h(P_4, S_3) \tag{55}$$

$$h_4 = h_3 - \eta_{mn}(h_3 - h_{4,s}) \tag{56}$$

Using the properties obtained at this point, the states at the suction nozzle, mixing section, and diffuser are then calculated in sequence:

*Table*4 *: Governing thermodynamic equations of the ejector*

Suction nozzle	$P_{10} = P_9' - \Delta P$ $h_{10,s} = h(P_{10}, S_9')$ $h_{10} = h_9' - \eta_{sn}(h_9' - h_{10,s})$ $c_{10} = \sqrt{2(h_9' - h_{10,s})}$
Mixing section	$P_5 = P_4 = P_{10}$ $c_5 = \sqrt{\eta_{ms}}\left(\frac{1}{1+\mu}c_4 + \frac{\mu}{1+\mu}c_{10}\right)$ $h_5 = \frac{1}{1+\mu}\left(h_4 + \frac{c_4^2}{2}\right) + \frac{\mu}{1+\mu}\left(h_{10} + \frac{c_{10}^2}{2}\right) - \frac{c_5^2}{2}$ $S_5 = S(P_5, h_5)$
Diffuser outlet	$h_6 = h_5 + \frac{c_5^2}{2}$ $h_{6,s} = h_5 + \eta_d(h_6 - h_5)$ $P_6 = P(h_{6,s}, S_5)$

Separator

At this stage, the properties have been obtained up to state point 6, immediately upstream of the separator. The separator outlet states are then determined as follows:

$$h_1 = h(P_6, x = 1) \tag{57}$$

$$h_7 = h(P_6, x = 0) \tag{58}$$

According to the stated assumptions, the separator is treated as having negligible pressure drop. Therefore:

$$P_6 = P_1 = P_7 \tag{59}$$

Compressor

The thermodynamic equations governing the compressor, following Ref. [1], are written as follows:

$$h_{2,s} = h(P_2, S_1) \tag{60}$$

$$h_2 = h_1 + \frac{h_{2,s} - h_1}{\eta_{comp}} \tag{61}$$

$$\eta_{comp} = 0.874 - 0.0135\frac{P_c}{P_e} \tag{62}$$

Heat Exchanger

Using the equation adopted for heat-exchanger effectiveness, the following relations are obtained:

$$\eta = \frac{T_7 - T_7'} {T_7 - T_9} \tag{63}$$

$$\rightarrow \quad T_7' = T_7 - \eta(T_7 - T_9) \tag{64}$$

Pressure losses in the heat exchanger are assumed negligible, so the pressure at each inlet is equal to that at its corresponding outlet. Therefore:

$$P_7' = P_7 \tag{65}$$

At this point, two independent thermodynamic properties have been determined for the required state, so the fluid state is fully specified.$7'$

Throttling Valve

Across the throttling valve, the enthalpy before and after the valve is assumed to be equal; the principal change is the pressure drop. Because the pressure drop inside the evaporator is also neglected, the corresponding pressure relation applies. Therefore:$h_8 = h_7{}' P_9 = P_8$

$$h_8 = h_7' \tag{66}$$

$$P_8 = P_9 \tag{67}$$

With the two independent properties obtained above, all thermodynamic properties at state point 8 can then be determined.

2-3-4 Convergence Analysis

2-3-4-1 First Convergence ConditionT_9'

By assuming the required temperature in the preceding equations, the fluid properties at all cycle state points can be evaluated. The first convergence condition is then checked using the energy balance across the heat recuperator.$T_9{}'$

$$h_7 - h_7' = h_9' - h_9$$

$$\rightarrow h_7' = h_7 - h_9' + h_9$$

As noted previously, pressure drop within the heat exchanger is neglected.

$$P_9' = P_9$$

The properties at the corresponding state point are then recalculated, giving:$9'$

$$T_9' = T(P = P_9', h = h_7{}')$$

The temperature calculated from the relation above is compared with the initially guessed temperature. If the difference between these two temperatures is smaller than the prescribed error

tolerance—0.0001 in this project—the temperature guess is accepted and the first convergence condition is satisfied.$T_9'T_9'$

$$if: \left|T_{9_{Guessed}}^{'} - T_{9_{Calculated}}^{'}\right| < error \quad \rightarrow \quad T_{9_{Guessed}}^{'} \ has\ Correct\ value!$$
$$else: T_{9_{Guessed}}^{'}\ Should\ be\ change!$$

2-3-4-2 Second Convergence Conditionμ

The second convergence condition is identical to that used for the cycle without a heat recuperator. An updated value of the relevant mass-flow-rate ratio is calculated from the governing relations and compared with the previously assumed value. If the difference is smaller than the prescribed error tolerance of 0.0001, the result is accepted and the initial assumption is considered valid. Otherwise, the value is updated iteratively until the following convergence condition is satisfied:$\frac{1}{1+\mu}x_6x'_6x_6x_6x'_6$-$x_6\mu\mu$

Convergence Criterion for the Project:

$$if: |\mu - \mu'| < error \quad \rightarrow \quad \mu\ has\ Correct\ value!$$
$$else: \mu\ Should\ be\ change!$$

This iterative check must therefore be performed for every set of cycle inputs in order to obtain the correct system solution.

2-3-4-3 Procedure for Applying the Convergence Conditions

The convergence procedure implemented in the project code extends the convergence condition described in the previous section. In addition to checking convergence of the mass-flow-related variable, the additional temperature convergence condition introduced by the heat recuperator must also be satisfied.$\mu T_9^{'}$

The convergence procedure begins with an initial step size of 10. If the initial guess is incorrect, the following cases are considered:

Case 1: The initial guess is greater than the value calculated from the governing equations.

In this case, the assumed value is reduced by the current step size.

Case 2: The initial guess is smaller than the value calculated from the governing equations.

In this case, the assumed value is increased by the current step size.

The step size used in the solution is then divided by 10.

The procedure continues until the difference between the assumed and calculated values becomes smaller than the prescribed error tolerance.

The same general procedure is used for both convergence variables. The correction step used for the temperature variable is necessarily different from that used for the mass-flow-ratio variable. The temperature-correction step is defined outside the inner iteration loop and remains fixed at that level, whereas the step used for the mass-flow-ratio variable must be redefined and progressively adjusted within each inner loop.$T_9'\mu\mu T_9'\mu T_9'$

2-4 Determination of the Best Cycle Operating Point

The first objective in this section is to obtain a continuous input-output relationship for the cycle. The variables considered are condenser temperature, evaporator temperature, and the pressure difference between state points 4 and 9, i.e., the pressure difference between the primary-nozzle outlet

and the evaporator outlet. A regression algorithm is first trained to predict COP for any combination of these inputs. An optimization algorithm is then applied to the regression model to identify the operating conditions that maximize COP. The complete procedure is discussed step by step below.ΔP

2-4-1 Definition of Initial Values and Upper and Lower Bounds for Optimization

In this section, data are generated for the vapor-compression refrigeration cycle configuration that includes the heat recuperator. Specific ranges are defined for condenser temperature, evaporator temperature, and the pressure difference between state points 9 and 4. These ranges follow the recommendations of Ref. [1].

The variable inputs are defined as follows:

$$\begin{cases} T_e = -5℃, +5℃ \\ T_c = 30℃, +50℃ \\ \Delta P = 0.5kPa - 50kPa \end{cases} \tag{68}$$

The constant inputs for each cycle are:

$$\begin{gathered} \eta_{mn} = 0.85 \\ \eta_{ms} = 0.95 \\ \eta_{sn} = 0.85 \\ \eta_{d} = 0.85 \\ \eta_{HE} = 0.80 \end{gathered} \tag{69}$$

A step size of 0.5 is used for the parameter variations. This gives 21 possible evaporator-temperature values, 41 condenser-temperature values, and 100 intermediate-pressure-difference values. By the multiplication principle, the complete grid therefore contains 86,100 data points for model training. The large training data set is intended to improve the quality of the fitted model.

2-4-2 Training the Regression Model in MATLAB

2-4-2-1 Model-Creation Procedure

16. Data Preparation and Import

The data are first imported as a table, after which the predictor variables and response variable are separated. This prepares the data for use in the regression model:

```
inputTable = trainingData;
predictorNames = {'delta_p', 'T_C', 'T_E'};
predictors = inputTable(:, predictorNames);
response = inputTable.COP;
isCategoricalPredictor = [false, false, false];
```

Here, the input table trainingData is copied into inputTable for use in the subsequent steps. predictorNames contains the names of the input variables. The corresponding columns are extracted from the table and stored in predictors. The response variable is the COP column. isCategoricalPredictor is a logical array indicating that none of the predictor variables is categorical.

17. Regression-Model Training

This step focuses on training a regression-tree model using the input data. The procedure includes several key stages intended to provide sufficient accuracy and generalization. The final objective is to obtain a regression model capable of accurately predicting COP from the input variables delta_p, T_C, and T_E.

```
regressionTree = fitrtree(predictors,response,'MinLeafSize', 4, 'Surrogate', 'off');
```

The fitrtree function is used to train the model. It fits a regression decision tree using the predictor data and response data. MinLeafSize is set to 4, meaning that each terminal leaf must contain at least four observations; this setting helps reduce overfitting. The Surrogate option is disabled, so surrogate split variables are not used.

18. Creating the Output Structure and Prediction Function

After training the model, a structure containing both the trained model and its prediction function is created so that the model can conveniently be used for new predictions:

```
predictorExtractionFcn = @(t) t(:, predictorNames);
treePredictFcn = @(x) predict(regressionTree, x);
trainedModel.predictFcn = @(x) treePredictFcn(predictorExtractionFcn(x));
```

Several anonymous functions are defined here. predictorExtractionFcn extracts the predictor columns identified by predictorNames from an input table. treePredictFcn uses the trained regression tree to calculate predicted values. Finally, trainedModel.predictFcn combines these operations and is used to predict COP for new input data.

19. Model Validation

The final step validates the model using cross-validation. This is used to assess predictive accuracy and the ability of the model to generalize to unseen data:

```
partitionedModel = crossval(trainedModel.RegressionTree, 'KFold', 5);
validationPredictions = kfoldPredict(partitionedModel);
validationRMSE = sqrt(kfoldLoss(partitionedModel, 'LossFun', 'mse'));
```

2-4-2-2 Summary

This MATLAB code trains a regression-tree model to predict COP from the input variables delta_p, T_C, and T_E. The workflow includes data preparation, model training, construction of a prediction function, and evaluation of predictive accuracy. The trained model can then predict COP for new operating conditions. Cross-validation is used to evaluate accuracy and generalization so that the suitability of the model for predicting new data can be assessed.

2-4-3 Optimization of Ejector Performance Using a Genetic Algorithm (GA) in MATLAB

The objective of this section is to optimize ejector-cycle performance using the model trained in the previous step. The regression-tree model developed to predict COP from delta_p, T_C, and T_E is used as a surrogate model, and a genetic algorithm (GA) is applied to search for the operating conditions that produce the maximum COP.

2-4-3-1 Optimization Procedure

The optimization consists of several stages: data preparation, model training, definition of variable bounds, configuration of the genetic algorithm, definition of the objective function, and execution of the optimization. Each stage is described separately below.

The data are first imported from an Excel file into MATLAB and assigned as the training data set.

1- Data Preparation

```
data = readtable('Book1.xlsx');
trainingData = data;
```

The Excel file Book1.xlsx is read and stored in a variable named data. The data are then assigned to trainingData for use during model training.

2- Model Training

The regression-tree model is trained using the training data prepared in the previous step.

```
[trainedModel, validationRMSE] = trainRegressionModel(trainingData);

disp(['Validation RMSE: ', num2str(validationRMSE)]);
```

The trainRegressionModel function defined earlier is called to train the regression tree. The function also calculates the validation root-mean-square error (RMSE). The RMSE value is then displayed as a measure of model accuracy.

3- Definition of Bounds

For the optimization, lower and upper bounds are specified for the input variables delta_p, T_C, and T_E.

```
lb = [0.5, 30+273.15,-5+273.15];
ub = [50, 50+273.15, +5+273.15];
```

The variables lb and ub contain the lower and upper bounds, respectively. The temperature limits are expressed in kelvin to remain consistent with the original data. For example, the lower bound is 0.5 for delta_p, 303.15 K for T_C, and 268.15 K for T_E. The upper limits are defined similarly.

4- Genetic-Algorithm Options

The genetic-algorithm parameters are configured to control the optimization procedure.

```
nvars = 3;
options = optimoptions('ga', 'Display', 'iter');
```

The variable nvars is set to 3 because three optimization variables are used. The GA options are then defined. Display is set to iter so that information from each optimization iteration is shown in the MATLAB console.

5- Definition of the Objective Function

An objective function is defined that predicts COP using the trained model and returns its negative value.

```
function negCOP = objectiveFunction(x, trainedModel)
    newData = table(x(1), x(2), x(3), 'VariableNames', {'delta_p', 'T_C', 'T_E'});
    predictedCOP = trainedModel.predictFcn(newData);
```

```
    negCOP = -predictedCOP;
end
```

The function objectiveFunction receives a vector x containing delta_p, T_C, and T_E. These values are converted into a table compatible with the trained regression model. COP is then predicted using the model's predictFcn. Because MATLAB's genetic algorithm performs minimization by default, the negative of the predicted COP is returned so that minimizing the objective corresponds to maximizing COP.

6- Running the Optimization

The genetic algorithm is executed to identify the input values that produce the maximum COP.

```
[optimalX, maxCOPNeg] = ga(@(x) objectiveFunction(x, trainedModel), nvars, [], [], [], [], lb, ub, [], options);
maxCOP = -maxCOPNeg;
disp('Optimal values:');
disp(['delta_p: ', num2str(optimalX(1))]);
disp(['T_C: ', num2str(optimalX(2))]);
disp(['T_E: ', num2str(optimalX(3))]);
disp(['Max COP: ', num2str(maxCOP)]);
```

The ga function executes the genetic algorithm. Its inputs include the objective function, number of variables, lower and upper bounds, and algorithm options. The output vector optimalX contains the optimum values of delta_p, T_C, and T_E, while maxCOPNeg is the negative of the maximum predicted COP. Multiplying maxCOPNeg by -1 gives the maximum COP. The optimum variables and maximum COP are then displayed.

2-4-3-2 Summary

Using the trained regression-tree model, ejector-cycle performance is optimized with the objective of maximizing COP. A genetic algorithm is used because it can search a complex parameter space without requiring derivatives. The workflow consists of data preparation, model training, definition of bounds and algorithm options, construction of the objective function, and execution of the optimization. The resulting optimum input values and maximum COP are then reported.

2-5 Ejector Fabrication and Design Procedure

2-5-1 Initial Assumptions

Because the internal two-phase flow regime of the ejector is complex to simulate and analytical relationships are not available for every geometric section, the design procedure uses empirical charts and correlations derived from experimental studies of two-phase ejectors [10].

Thermodynamic ejector models are commonly formulated using two principal mixing assumptions, CAM and CPM [10]. In the Constant-Area Mixing (CAM) approach, the primary and secondary streams mix in a constant-cross-sectional-area section and the pressure of the combined flow increases as mixing proceeds. In the Constant-Pressure Mixing (CPM) approach, the geometry allows the two streams to mix at approximately constant pressure, so the pressure remains essentially unchanged throughout the mixing process.

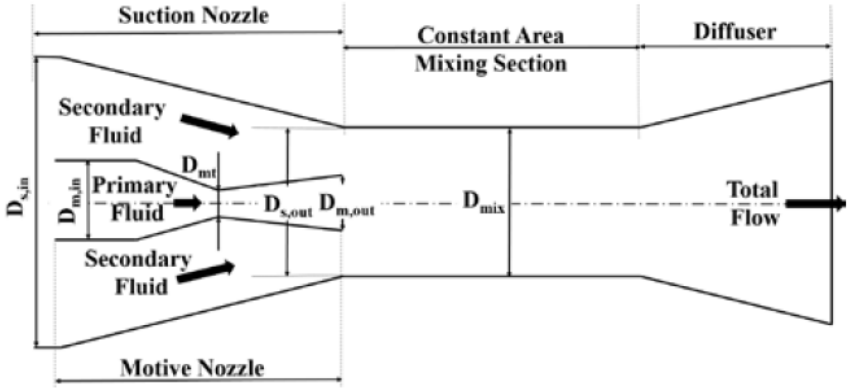


Figure 13: Geometric characteristics of the ejector reported in Ref. [10]

In this section, the ejector analysis and fabrication procedure is developed under the following initial assumptions:

$$T_c = -5°C$$

$$D_{s_{in}}(Suction\ Nozzle\ inlet\ Diameter): 60\ mm$$

$$D_{m_{in}}(Motive\ Nozzle\ inlet\ Diameter): 15\ mm$$

2-5-2 Governing Equations

The governing equations are formulated using the relationships presented in Ref. [10]. In the Results chapter, the charts obtained from these equations are then used to complete the two-phase ejector design procedure.

The entrainment ratio, defined as the ratio of the secondary-fluid mass flow rate to the motive-fluid mass flow rate, is a critical ejector-performance parameter that should be maximized in a well-designed ejector. It is defined as follows:

$$\mu = \frac{\dot{m}_s}{\dot{m}_m} \tag{70}$$

The equations used to calculate the principal ejector dimensions are based on Ref. [9]. First, the secondary-stream mass flow rate is determined from the required cooling capacity and the enthalpy difference across the evaporator at the optimum suction-nozzle pressure for the operating temperatures under consideration. The primary-stream mass flow rate is then calculated from the optimum entrainment ratio. The primary and secondary mass flow rates are obtained as follows:

$$\dot{m}_s = \frac{\dot{Q}_e}{\Delta h_{e,opt}} \tag{71}$$

$$\dot{m}_m = \frac{\dot{m}_s}{w_{opt}} \tag{72}$$

Second, the throat pressure is determined iteratively. The speed of sound corresponding to an assumed throat pressure is compared with the velocity calculated from the energy equation. The iterations continue until the calculated flow velocity and the local speed of sound converge within an acceptable tolerance, because the Mach number at the throat must be equal to one. The speed of sound and void fraction for the homogeneous two-phase flow are defined as follows:$P_{mt}\frac{V_g}{V_I} = 1$

$$c = \left[\rho_{tp}\left(\frac{\alpha}{\rho_g c_g^2} + \frac{1-\alpha}{\rho_l c_l^2}\right)\right]^{-1/2} \quad (73)$$

$$\rho_{tp} = \alpha\rho_g + (1-\alpha)\rho_l \quad (74)$$

$$\alpha = \frac{X_m}{X_m + (1-X_m)\frac{\rho_g}{\rho_l}} \quad (75)$$

In the equation above, the corresponding symbol denotes vapor quality.X_m

The throat velocity calculated from the energy equation is given by:

$$h_{mt} = (1-\eta_m)h_{m,in} + \eta_m h(s_{m,in}, P_{mt}) \quad (76)$$

$$u_{mt} = \sqrt{2(h_{m,in} - h_{mt}) + u_{m,in}^2} \quad (77)$$

Overall, the ejector geometry is calculated using an iterative procedure. The convergence criterion is that the Mach number at the motive-nozzle throat be equal to one. An initial value is first assumed for the throat pressure. The two-phase mixture density and void fraction are then calculated, followed by the local speed of sound at the throat, c. Independently, the throat enthalpy h_mt is determined and the flow velocity u_mt is calculated from the energy equation. If comparison of these quantities gives a Mach number of unity, the iterative process is terminated and the converged values are used to calculate the required ejector dimensions.$P_{mt}\alpha$

After the throat pressure has been accurately estimated using the iterative procedure, the throat diameter is calculated from the following relation:

$$D_{mt} = \sqrt{\frac{4\dot{m}_m}{\pi\rho_{mt}u_{mt}}} \quad (78)$$

The motive-nozzle outlet diameter and suction-nozzle outlet diameter are calculated from the optimum expansion pressure at their outlets. The diameter of the constant-area mixing section is taken to be equal to the suction-nozzle outlet diameter.

1- Motive-Nozzle Outlet Diameter:

$$D_{m,\,\text{out}} = \sqrt{\frac{4\dot{m}_m}{\pi\rho_{m,\,\text{out}}\,u_{m,\,\text{out}}}} \quad (79)$$

2- Suction-Nozzle Outlet Diameter:

$$D_{s,\,\text{out}} = \sqrt{\frac{4\dot{m}_s}{\pi\rho_{s,\,\text{out}}\,u_{s,\,\text{out}}} + D_{m,\,\text{out}}^2} \quad (80)$$

3- Constant-Area Mixing-Section Diameter:

$$D_{mix} = \sqrt{\frac{4(\dot{m}_m + \dot{m}_s)}{\pi \rho_{mix,out} u_{mix,out}}} \tag{81}$$

Using the equations derived in this section together with the recommended optimum angles for the different ejector sections reported in Ref. [11], the ejector can then be analyzed and dimensioned for fabrication.

Chapter 3: Results

3-1 Results of the Two-Phase Ejector Vapor-Compression Refrigeration Cycle without a Heat Recuperator

3-1-1 Simulation Conditions

The thermodynamic cycle in this project was simulated in MATLAB using the REFPROP interface. The reasons for selecting this environment instead of EES were discussed in detail in Chapter 2.

The working fluid used in this project is R1234yf. It was selected in part because of its thermodynamic similarity to R134a. The main reason for not using R134a directly, however, was the need to validate the simulation results. The reference studies examined in this project, including Refs. [1-3], used R1234yf. Therefore, the same refrigerant was adopted so that the present calculations could be compared with published results and the corresponding simulation error could be quantified.

The simulation inputs are divided into two groups: fixed inputs, which include the efficiencies assigned to the different cycle components, and variable inputs, which include condenser temperature, evaporator temperature, and the pressure difference between state points 4 and 9, i.e., between the primary-nozzle outlet and the evaporator outlet.ΔP

Accordingly, the variable inputs are:

$$\begin{cases} T_e = -5℃, +5℃ \\ T_c = 30℃, +50℃ \\ \Delta P = 0.5kPa - 50kPa \end{cases}$$

The fixed inputs for each cycle are:

$$\eta_{mn} = 0.85$$
$$\eta_{ms} = 0.95$$
$$\eta_{sn} = 0.85$$
$$\eta_{d} = 0.85$$

To evaluate the effect of each variable input, two of the three variables are held constant while the third is varied over its prescribed range, and the resulting COP is calculated.

3-1-1-1 Case 1: Calculation of COP with Variable Pressure DifferenceΔP

1- Problem Inputs

In this case, the condenser and evaporator temperatures are held constant, while the pressure difference between state points 4 and 9 is varied.ΔP

$$\begin{cases} T_e = 5℃ \equiv 278.15K \\ T_c = 40℃ = 313.15K \\ \Delta P: 0.5kPa - 50kPa \end{cases}$$

The fixed inputs are as follows:

$$\eta_{mn} = 0.85$$
$$\eta_{ms} = 0.95$$
$$\eta_{sn} = 0.85$$
$$\eta_{d} = 0.85$$

2- Simulation Outputs

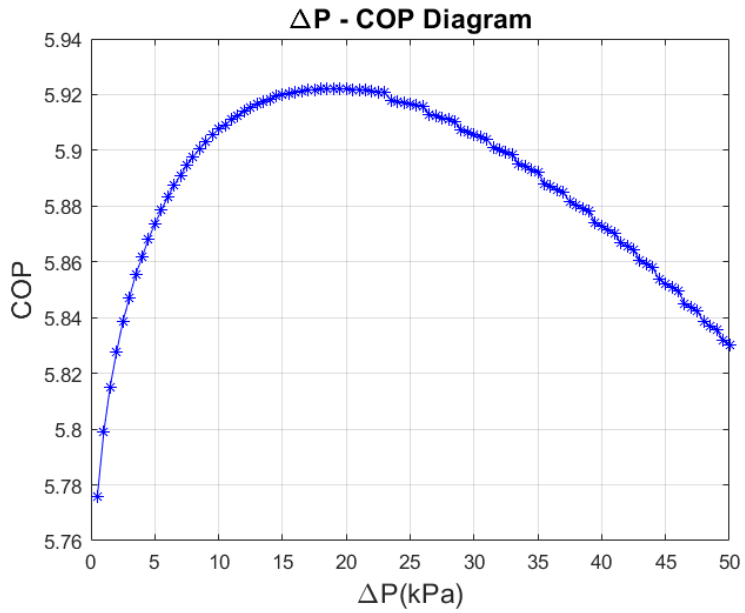


Figure 14: COP versus pressure variation for the cycle without a heat recuperatorΔP

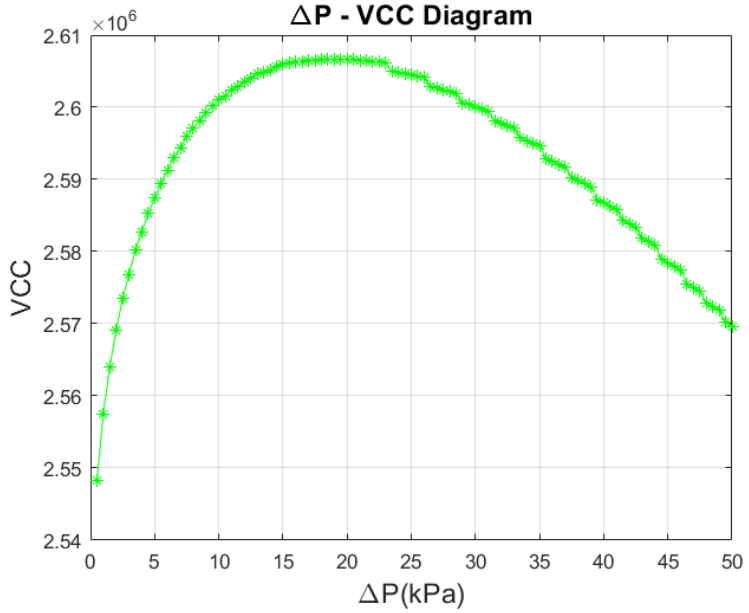


Figure 15: VCC versus pressure difference ΔP for the cycle without a heat recuperator

3-1-1-2 Case 2: Calculation of COP with Variable Condenser TemperatureT_c

1- Problem Inputs

In this case, the pressure difference between state points 4 and 9 and the evaporator temperature are held constant, while the condenser temperature is varied.ΔP

$$\begin{cases} T_e = +5℃ \\ \Delta P = 14kPa \\ T_c = 30℃, +50℃ \end{cases}$$

2- Simulation Outputs

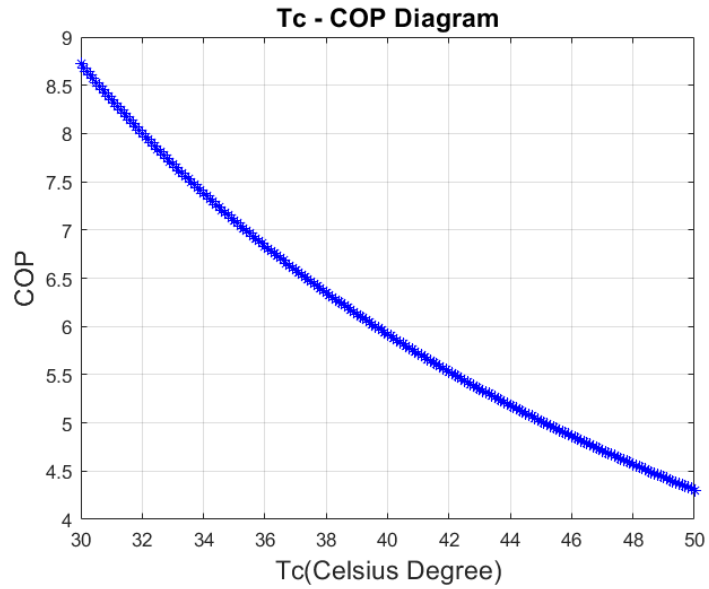


Figure 16: COP versus condenser-temperature variation for the cycle without a heat recuperator

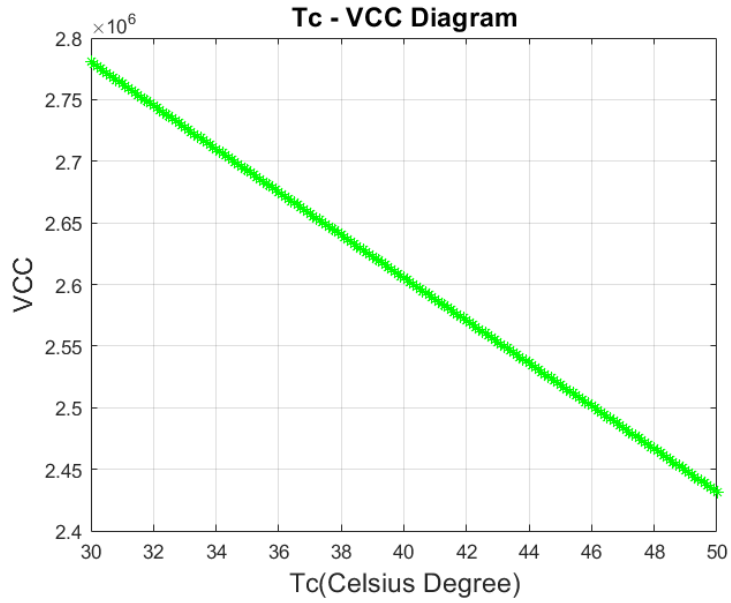


Figure 17: VCC versus condenser-temperature variation for the cycle without a heat recuperator

3-1-1-3 Case 3: Calculation of COP with Variable Evaporator TemperatureT_e

1- Problem Inputs

In this case, the pressure difference between state points 4 and 9 and the condenser temperature are held constant, while the evaporator temperature is varied.ΔP

$$\begin{cases} T_e = 5℃ \equiv 278.15K \\ T_c = 40℃ = 313.15K \\ \Delta P: 0.5kPa - 50kPa \end{cases}$$

2- Simulation Outputs

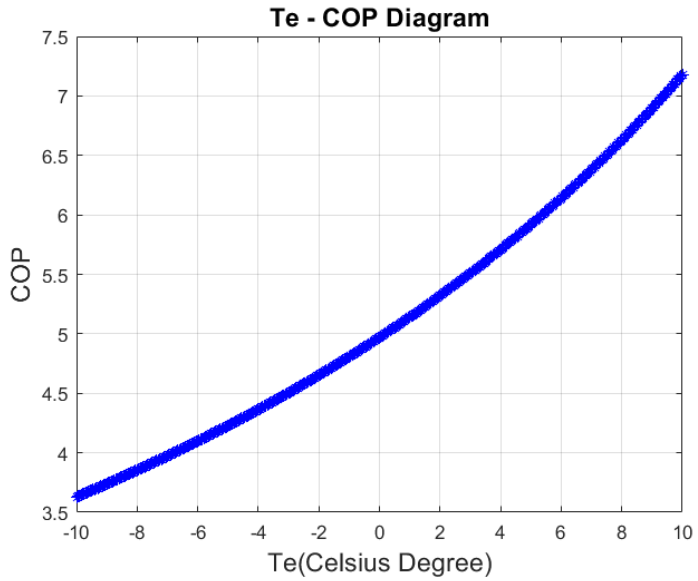


Figure 18: COP versus evaporator-temperature variation for the cycle without a heat recuperator

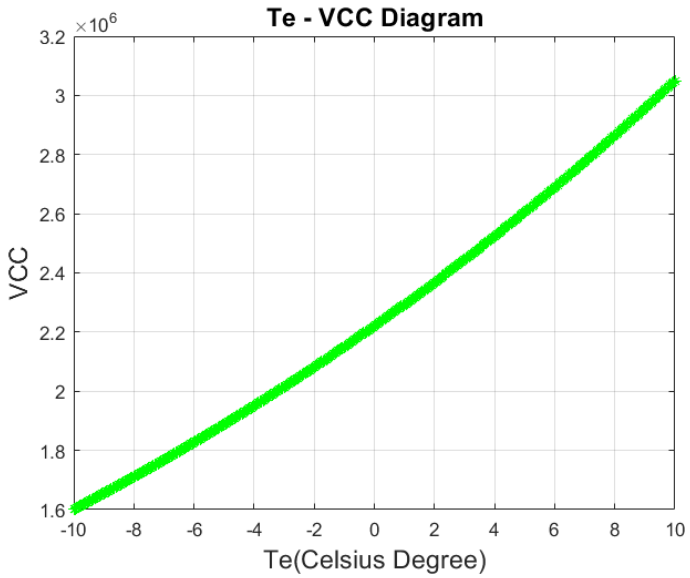


Figure 19: VCC versus evaporator-temperature variation for the cycle without a heat recuperator

3-1-2 Validation of the Results

Any scientific analysis requires its calculated outputs to be validated against a reliable reference.

For this purpose, the present simulation outputs are compared with the corresponding results reported in Ref. [1], and the relative error is subsequently calculated.

The validation procedure uses a graph-data extraction tool to digitize the values plotted in the reference article. The extracted data are then imported into Microsoft Excel for comparison.

3-1-3 Validation Based on Comparison of COP Values

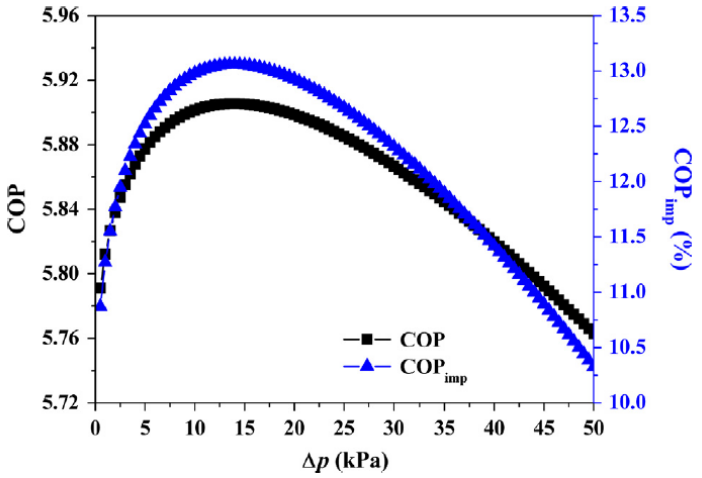


Figure 20: COP variation for the cycle without a heat recuperator based on Ref. [1]ΔP

The graph above is reproduced from Ref. [1]. The values from this graph are compared with the corresponding results of the present simulation obtained by varying the pressure difference while keeping the condenser and evaporator temperatures fixed. The relative error between corresponding data points is then calculated.ΔP

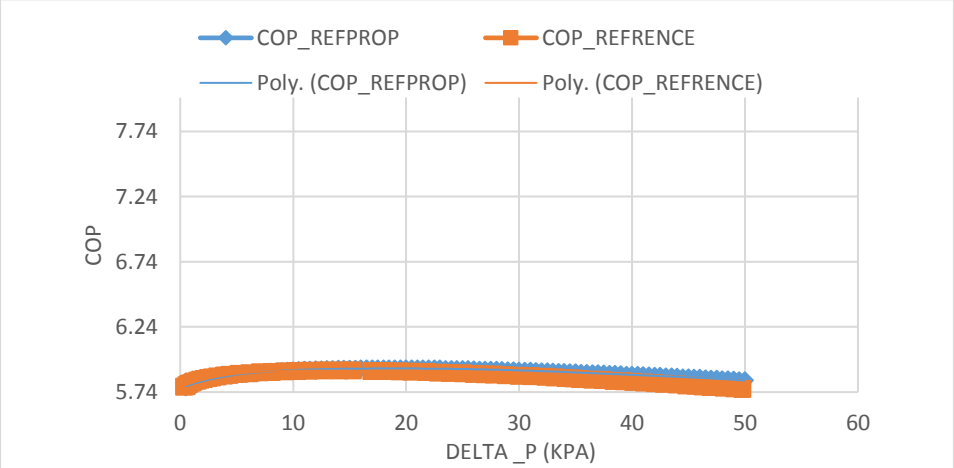


Figure 21: Comparison between COP values calculated in the present simulation and data reported in Ref. [1]

In the graph above, the reference curve represents the COP values reported in the cited study and is treated as the benchmark. The second curve represents the COP values calculated using the REFPROP-based simulation developed in this project. The two curves are very close and overlap to a good approximation. To quantify the remaining discrepancy, the relative error is calculated at each data point.

$$Relative\ Error = \left|\frac{Refrence\ Value - Refprop\ Value}{Refrence\ Value}\right| \times 100 \quad (82)$$

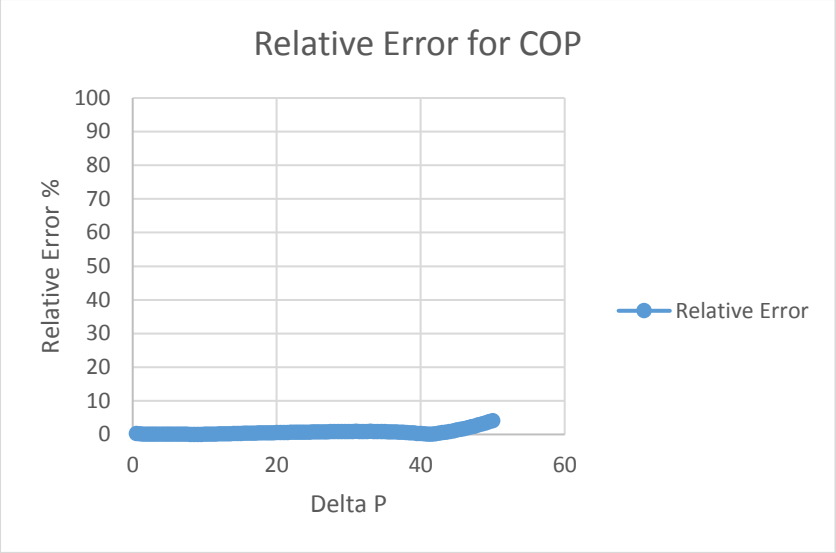


Figure 22: Relative-error diagram for the calculated COP

The maximum relative error is approximately 4%. Moreover, at about 90% of the evaluated points, the relative error is below 1%. These results indicate good agreement between the present cycle simulation and the reference data, supporting the use of the same simulation procedure for the remaining analyses.

3-2 Effect of Adding a Heat Recuperator on COP

The following figure shows the variation of cycle COP with condenser temperature for the configuration including a heat recuperator.

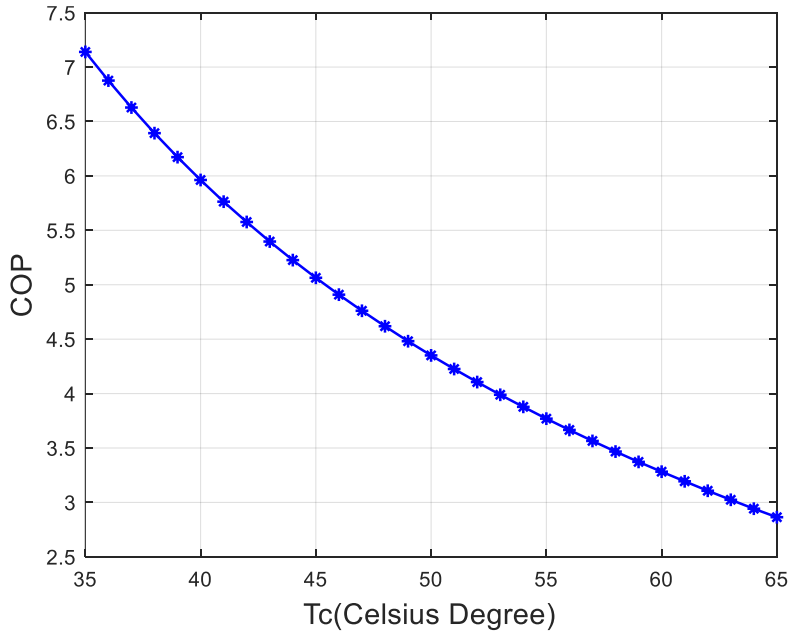


Figure 23: Variation of cycle COP with condenser temperature when a heat recuperator is included

The next figure shows the difference in cycle COP between the configurations with and without a heat recuperator as a function of condenser temperature.

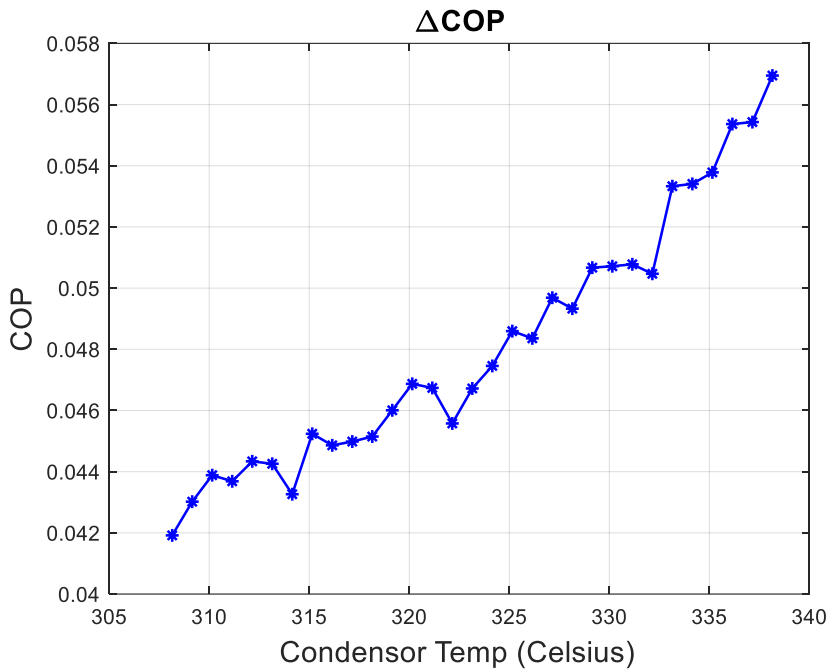


Figure 24: Difference in cycle COP between configurations with and without a heat recuperator as a function of condenser temperature

As can be seen, adding the heat recuperator has only a limited effect on cycle COP. For a clearer comparison, the following figure presents the percentage change in COP caused by the heat recuperator as a function of condenser temperature.

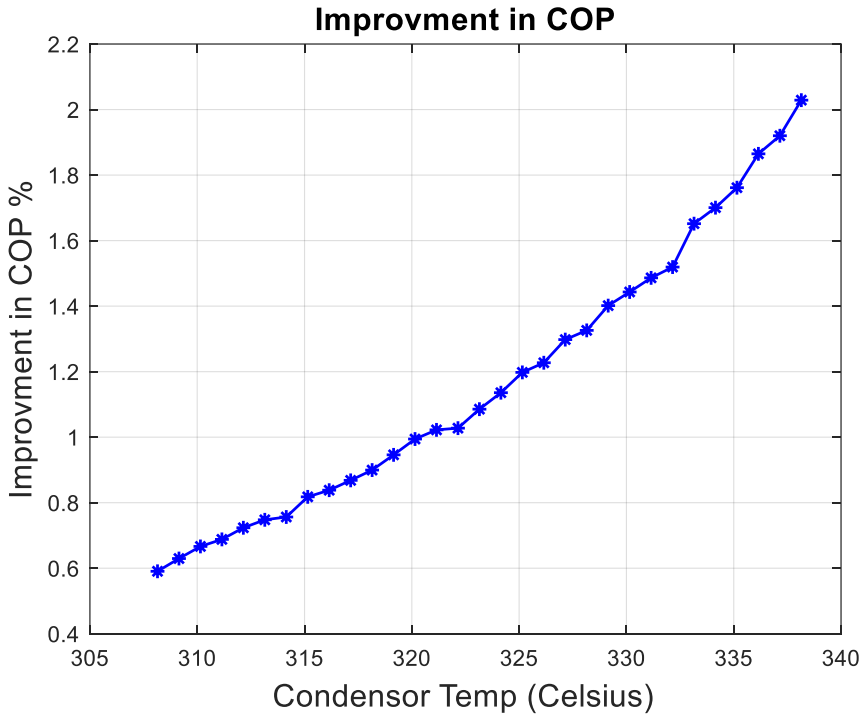


Figure 25: Percentage change in cycle COP due to the heat recuperator as a function of condenser temperature

As shown, the maximum change in cycle COP caused by the heat recuperator is approximately 2%. Because this improvement is small, the heat recuperator is not retained in the ejector-expansion vapor-compression refrigeration cycle considered in the subsequent design.

3-3 Optimization Results for the Two-Phase Ejector Vapor-Compression Refrigeration Cycle

As shown in the previous section, adding the heat exchanger produces only a small increase in cycle COP, reaching about 2% at best. Therefore, its use is not considered economically justified from the Design for Environment (DFE) perspective adopted in this project, because the environmental and economic costs associated with manufacturing and using an additional heat exchanger are judged to outweigh such a limited COP improvement.

Accordingly, the optimization is carried out using data from the ejector-based vapor-compression refrigeration cycle without the heat recuperator. The objective is to determine the operating point at which COP is maximized. The optimization methodology and the implemented code were described in detail in the Methodology chapter; this section therefore focuses on the resulting outputs.

3-3-1 Generation of a Finite Data Set from the Modeled Cycle

Ranges of condenser temperature, evaporator temperature, and the intermediate pressure difference between state points 4 and 9 are first specified as model inputs. The corresponding COP values are then generated for all combinations of these inputs.ΔP

This data-generation process is implemented using three nested for-loops.

The variable inputs are:

$$\begin{cases} T_e = -5℃, +5℃ \\ T_c = 30℃, +50℃ \\ \Delta P = 0.5kPa - 50kPa \end{cases} \quad (83)$$

The fixed inputs for each cycle are:

$$\begin{aligned} \eta_{mn} &= 0.85 \\ \eta_{ms} &= 0.95 \\ \eta_{sn} &= 0.85 \\ \eta_d &= 0.85 \\ \eta_{HE} &= 0.80 \end{aligned} \quad (84)$$

The following figure shows a small portion of the cycle data generated for training the regression model.

A1 | delta_p

	A	B	C	D	E
1	delta_p	COP	T_C	T_E	
2	0.5	8.578334	303.15	278.15	
3	1	8.607725	303.15	278.15	
4	1.5	8.628884	303.15	278.15	
5	2	8.644277	303.15	278.15	
6	2.5	8.657273	303.15	278.15	
7	3	8.668566	303.15	278.15	
8	3.5	8.677552	303.15	278.15	
9	4	8.685514	303.15	278.15	
10	4.5	8.691623	303.15	278.15	

Figure 26: A portion of the cleaned data used in the regression algorithm

Reading Data from the Excel File:

data = readtable('Book1.xlsx');

In this line of code, data are read from the Excel file Book1.xlsx and stored in a variable named data. MATLAB's readtable function imports the spreadsheet as a table that can contain multiple rows and columns.

3-3-2 Calling the Function Obtained from the Regression Algorithm

3-3-2-1 How to Call the Function

The previous section described the code used to train the regression model. In this section, the trained predictor is used to estimate COP at any desired point. The prediction function receives delta_p_new, representing the pressure difference between state points 4 and 9; T_C_new, representing condenser temperature; and T_E_new, representing evaporator temperature. The procedure is described in more detail below.

1- Assigning the Imported Data as Training Data

```
trainingData = data;
```

The data read from the Excel file are assigned to a variable named trainingData. This variable is used as the training data set for the regression model.

The imported Excel data are assigned to trainingData so that they can be used to train the regression model.

2- Training the Regression Model

```
[trainedModel, validationRMSE] = trainRegressionModel(trainingData);
```

The trainRegressionModel function is called to train the regression model using trainingData. The function returns two outputs: trainedModel, which contains the trained regression model, and validationRMSE, which contains the validation root-mean-square error.

3- Displaying the Validation Root-Mean-Square Error

```
disp(['Validation RMSE: ', num2str(validationRMSE)]);
```

The validationRMSE value is converted to text and displayed together with a descriptive label in the MATLAB Command Window.

4- Defining New Inputs for Prediction

```
delta_p_new = [35; 33];
T_C_new = [303.15; 320.15];
T_E_new = [279.15; 279.65];
```

These three lines define new values of delta_p_new, T_C_new, and T_E_new for COP prediction. Each array contains two input cases that will be passed to the trained model.

5- Creating the New Data Table

```
newData = table(delta_p_new, T_C_new, T_E_new, 'VariableNames', {'delta_p', 'T_C', 'T_E'});
```

The new input arrays delta_p_new, T_C_new, and T_E_new are converted into a table. The table function assigns the column names 'delta_p', 'T_C', and 'T_E' and creates a new table named newData.

6- Predicting COP Values Using the Trained Model

```
predictedCOP = trainedModel.predictFcn(newData);
```

The model's predictFcn is called with newData as the input. The output is a vector of predicted COP values stored in predictedCOP.

7- Displaying the Predicted COP Values

```
disp('Predicted COP values:');
disp(predictedCOP);
```

3-3-2-2 Evaluation of Prediction Accuracy Using Known Data

As an example, two sets of known input conditions are passed to the prediction function and the resulting predictions are compared with the corresponding cycle-simulation values.

First Input:

$$\begin{cases} T_e = 6°C \equiv 279.15 \\ T_c = 30°C \equiv 303.15 \\ \Delta P = 35kPa \end{cases}$$

Second Input:

$$\begin{cases} T_e = 6.5°C \equiv 279.65 \\ T_c = 47°C \equiv 320.15 \\ \Delta P = 35kPa \end{cases}$$

The input values above are placed in the input arrays as follows:

```
delta_p_new = [35; 33];
T_C_new = [303.15; 320.15];
T_E_new = [279.15; 279.65];
```

The desired inputs are then passed to the prediction function obtained from the regression model:

```
newData = table(delta_p_new, T_C_new, T_E_new, 'VariableNames', {'delta_p', 'T_C', 'T_E'});
predictedCOP = trainedModel.predictFcn(newData);
disp('Predicted COP values:');
disp(predictedCOP);
```

The following outputs are obtained:

```
Validation RMSE: 0.038594
Predicted COP values:
    8.9277
    4.9389
```

Figure27 : COP output predicted by the regression algorithm using the specified inputs

Table5 : Exact COP outputs based on cycle-simulation data obtained in REFPROP

$\Delta P(kPa)$	$Exact\ COP$	$T_c(K)$	$T_e(K)$
35	9.0006	303.15	279.15
33	4.95489	320.15	279.65

Relative-Error Calculation for the First Input

$$Relative\ Error = \frac{|Calculated\ COP - Exact\ COP|}{Exact\ COP} = \frac{|8.9277 - 9.0006|}{9.0006} \quad (85)$$
$$= 0.0081 \equiv 0.8\%$$

Relative-Error Calculation for the Second Input

$$Relative\ Error = \frac{|Calculated\ COP - Exact\ COP|}{Exact\ COP} = \frac{|4.9389 - 4.95489|}{4.9389} \quad (86)$$
$$= 0.00323 \equiv 0.323\%$$

As can be seen, the model obtained from the regression algorithm is highly accurate for the two test cases considered here, with relative errors below 1%.

3-3-3 Cycle-Performance Optimization Using a Genetic Algorithm (GA) in MATLAB

Using the regression function developed in the previous section, the objective is now to determine the optimum operating point of the cycle. The ejector will subsequently be designed using these optimum operating inputs so that the cycle can achieve the highest possible COP within the specified

ranges. The optimization code and its implementation are explained step by step below, together with the bounds applied to the input variables.

$$\begin{cases} T_e = -5℃, +5℃ \\ T_c = 30℃, +50℃ \\ \Delta P = 0.5kPa - 50kPa \end{cases} \quad (87)$$

3-3-3-1 Calling the Optimization Algorithm and Defining Its Initial Settings

The following code extends the regression-training code described in the previous section. Because the regression-model training procedure has already been explained in detail, only the optimization-related portion is discussed here.

1- Training the Regression Model

```
close all;
clc;
data = readtable('Book1.xlsx');
trainingData = data;
[trainedModel, validationRMSE] = trainRegressionModel(trainingData);
disp(['Validation RMSE: ', num2str(validationRMSE)]);
```

At this stage the regression model has been created. The next step is to define the allowable ranges of the variables for the optimization.

2- Defining the Variable Bounds

```
lb = [0.5, 30+273.15, -5+273.15];
ub = [50, 50+273.15, +5+273.15];
```

These lines define the allowable ranges for the input variables delta_p, T_C, and T_E. The variables lb and ub specify the lower and upper bounds, respectively. The bounds are defined as follows:

$$\begin{cases} T_e = -5℃, +5℃ \\ T_c = 30℃, +50℃ \\ \Delta P = 0.5kPa - 50kPa \end{cases}$$

3- Defining the Number of Variables

```
nvars = 3;
```

This line specifies that the genetic algorithm has three input variables.

4- Executing the Genetic Algorithm

```
options = optimoptions('ga', 'Display', 'iter');
[optimalX, maxCOPNeg] = ga(@(x) objectiveFunction(x, trainedModel), nvars, [], [], [], [], lb, ub, [], options);
```

Here, options defines the genetic-algorithm settings. The optimoptions function is used to configure the optimizer. In this case, only iteration-by-iteration output is enabled by setting ('Display', 'iter').

The ga function executes the genetic algorithm. Its inputs include the objective function, the number of variables nvars, optional linear constraints (left empty here), and the lower and upper bounds lb and ub.

The ga outputs include optimalX, which contains the optimum variable values, and maxCOPNeg, which contains the optimized negative COP value used by the minimization routine.

5- Calculating and Displaying the Maximum COP

```
maxCOP = -maxCOPNeg;
```

The negative COP returned by the genetic algorithm, maxCOPNeg, is converted back to a positive COP. Because the optimization objective was formulated as the negative of COP, this sign change yields the true maximum COP.

6- Displaying the Optimization Results

```
disp('Optimal values:');
disp(['delta_p: ', num2str(optimalX(1))]);
disp(['T_C: ', num2str(optimalX(2))]);
disp(['T_E: ', num2str(optimalX(3))]);
disp(['Max COP: ', num2str(maxCOP)]);
```

The optimum input variables stored in optimalX and the maximum COP stored in maxCOP are displayed in the MATLAB Command Window. Each disp statement prints one of the optimized quantities.

7- Definition of the Objective Function

```
function negCOP = objectiveFunction(x, trainedModel)
newData = table(x(1), x(2), x(3), 'VariableNames', {'delta_p', 'T_C', 'T_E'});
   predictedCOP = trainedModel.predictFcn(newData);
   negCOP = -predictedCOP;
end
```

In this section, the objective function is defined:

The input to objectiveFunction is a vector x containing the values of delta_p, T_C, and T_E.

These values are converted into a new data table, newData.

The trained model, trainedModel, is then used to predict COP.

Because the genetic algorithm is formulated as a minimization algorithm by default, the negative of COP (negCOP) is returned so that minimizing the objective function corresponds to maximizing COP.

3-3-3-2 Cycle-Performance Optimization Results

Optimization terminated: average change in the fitness value less than options.FunctionTolerance.

Optimal values:

delta_p: 0.5

T_C: 303.15

T_E: 268.15

Max COP: 8.68

The maximum COP obtainable from this cycle within the specified design space is therefore 8.68, corresponding to the following inputs:

$$\begin{cases} T_c = 303.15K = 30°C \\ T_e = 268.15K = -5°C \\ \Delta P = 0.5kPa \\ COP_{max} = 8.68 \end{cases} \quad (88)$$

3-4 Two-Phase Ejector Design Results

3-4-1 Calculation of the Dimensions and Angles of the Different Ejector Sections

The general ejector configuration considered for fabrication is shown below:

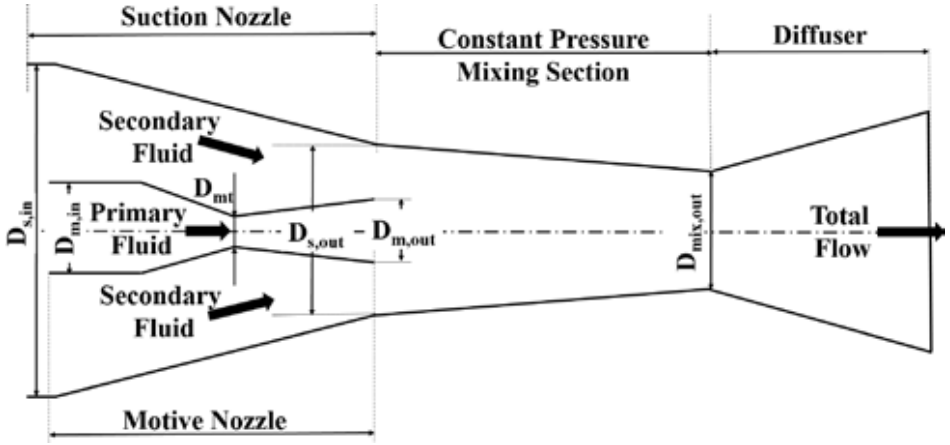


Figure 28: Two-phase ejector and subdivision of its different components [9]

With the optimum condenser temperature, evaporator temperature, and intermediate pressure difference now available, the design charts reported in Ref. [9] can be used. By examining each chart, dimensions of the ejector sections that could not be obtained directly from the analytical model are determined.ΔP

The calculations in this section use the assumptions listed below. The evaporator temperature is taken from the cycle-optimization results obtained in the preceding sections:

$$T_c = -5°C$$

$$D_{s_{in}}(Suction\ Nozzle\ inlet\ Diameter): 60\ mm$$

$$D_{m_{in}}(Motive\ Nozzle\ inlet\ Diameter): 15\ mm$$

The inlet diameters of the suction-nozzle and mixing sections are first specified according to the adopted design procedure, after which the remaining diameters are calculated. Once all section diameters are known, the corresponding section lengths can be determined from the recommended ejector angles.

The optimum values obtained from the cycle optimization are as follows:

$$\begin{cases} T_c = 303.15K = 30°C \\ T_e = 268.15K = -5°C \\ \Delta P = 0.5kPa \\ COP_{max} = 8.68 \end{cases}$$

3-4-1-1 Calculation of the Motive-Nozzle Throat and Outlet Diameters

For this chart, the only parameter required to determine the relevant diameters is the evaporator temperature. Therefore, using the optimized evaporator temperature gives:

$$T_e = 268.15K = -5°C$$

$$Fluid: R1234yf$$

$$\rightarrow \begin{cases} D_{mt}(Motive\ nozzle\ throat\ diameter) = 2.8mm \\ D_{m_{out}}(Motive\ nozzle\ outlet\ diameter) = 3.4mm \end{cases}$$

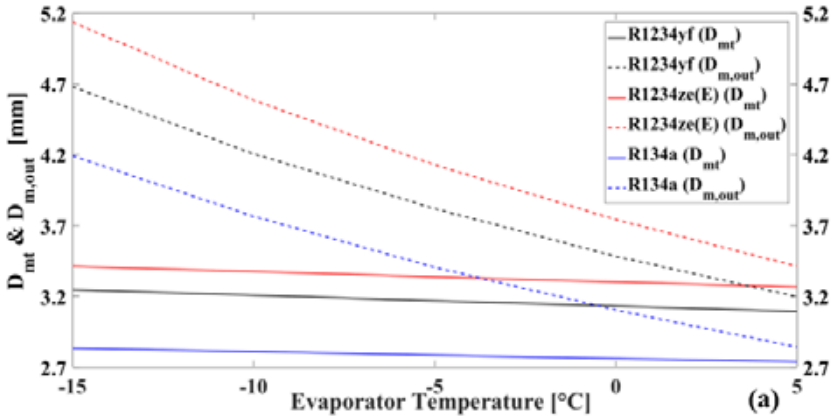


Figure 29: Motive-nozzle throat and outlet diameters as functions of evaporator temperature [9]

3-4-1-2 Calculation of the Mixing-Section Diameter

For this chart, the evaporator temperature is again the only parameter required to determine the relevant diameter. An important point indicated by the geometry is that the mixing-section inlet diameter is equal to the suction-nozzle outlet diameter. Therefore, the corresponding equality is applied.$D_{mix} = D_{s_{out}}$

Using the optimized evaporator temperature, the resulting value is:

$$T_e = 268.15K = -5°C$$

$$Fluid: R1234yf$$

$$\rightarrow D_{mix} = D_{S_{out}} = 8mm$$

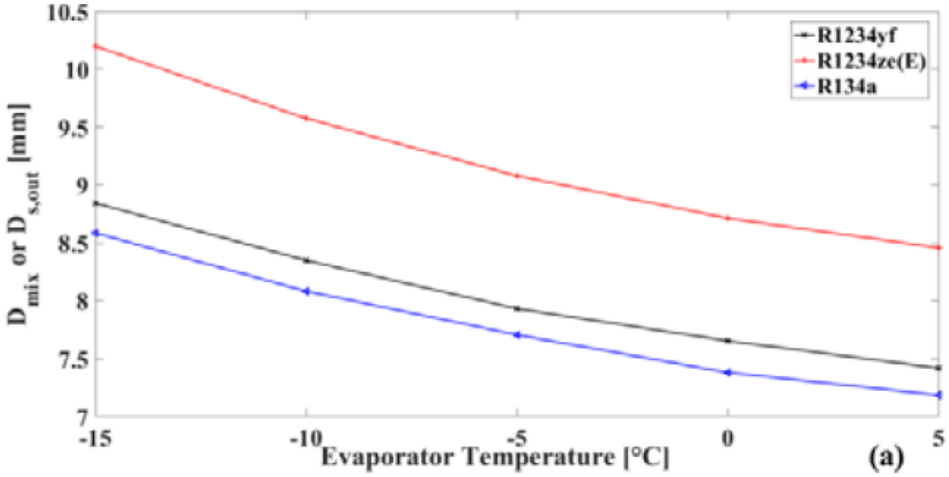


Figure 30: Mixing-section inlet diameter as a function of evaporator temperature [9]

3-4-1-3 Calculation of the Angle of Each Section

Based on the recommendations of the cited study, a 15° divergence angle is used for the diffuser and its length is taken as 59 mm. The mixing-section length is taken as 23.6 mm and its outlet diameter as 5.9 mm [11].$mmmm$

3-4-1-4 Summary of the Calculated Lengths and Angles of the Ejector Sections

Using the optimum operating values obtained from the preceding optimization, the lengths and angles of the different ejector sections were determined.

The resulting dimensions are summarized in the following table:

Table6 : Angles and lengths of the different ejector sections based on the specified inputs

Section lengthmm	Outlet diametermm	Inlet diametermm	Section name
30	3.4 Throat diameter: 2.8	15	**Motive nozzle**
40	8	60	**Suction nozzle**
23.6	5.9	8	**Mixing section**
59	21.5	5.9	**Diffuser**

3-4-2 Implementation of the Ejector Geometry in SolidWorks

Using the calculated lengths and angles of the different ejector sections, the final two-phase ejector geometry was modeled in SolidWorks. The resulting dimensions and angles are shown in the following figures.

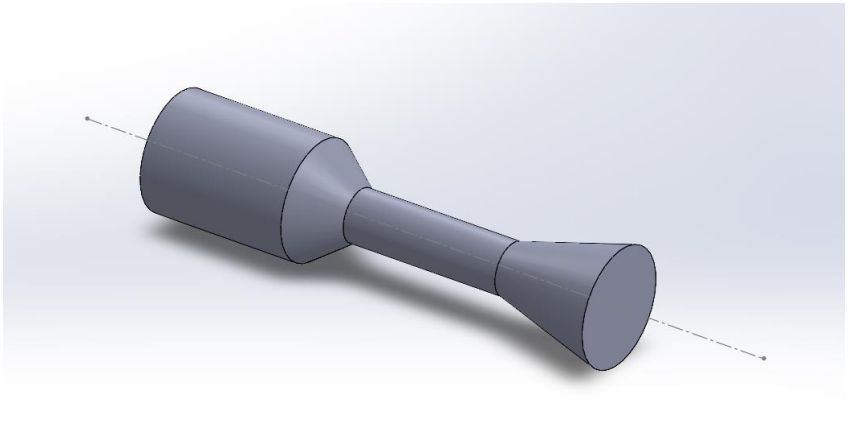


Figure 31: SolidWorks model of the ejector based on the lengths and angles obtained in the previous section

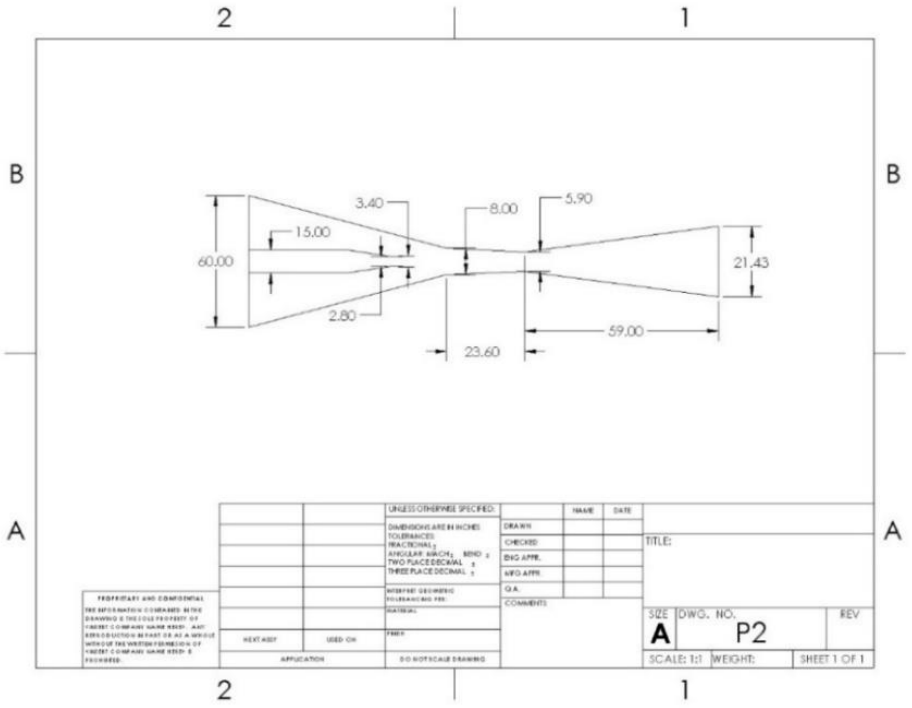


Figure 32: Front-view drawing with the specified dimensions of the two-phase ejector created in SolidWorks

3-4-2-1 Three-View Drawing of the Ejector Modeled in SolidWorks

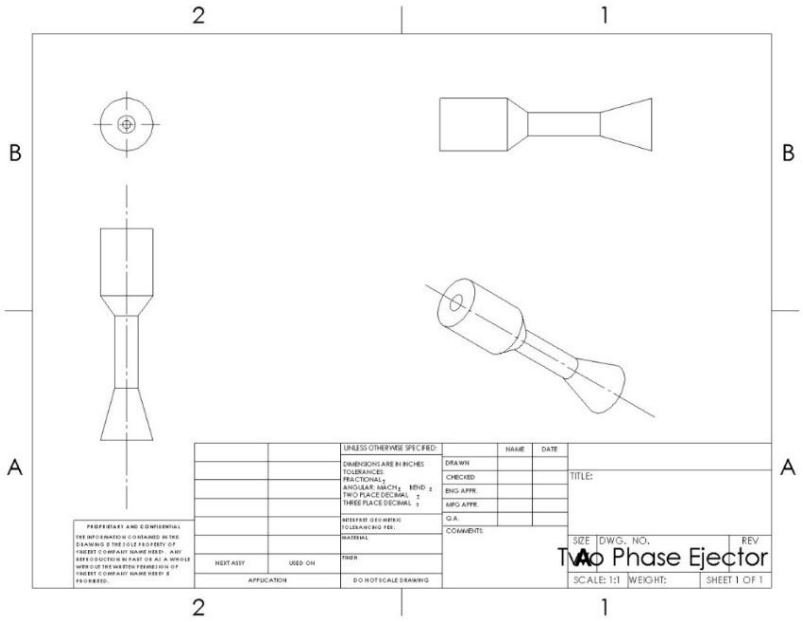


Figure 33: Three-view drawing of the designed ejector created in SolidWorks

Chapter 4: Summary and Conclusions

The main objective of this project was to improve the performance of vapor-compression refrigeration systems and reduce energy consumption. For this purpose, an ejector-based vapor-compression refrigeration cycle was investigated. By recovering part of the expansion-related pressure energy, the ejector can improve cycle performance and increase COP. The study used Refs. [1], [2], and [10] together with REFPROP-based simulation and produced the results summarized in this chapter.

The ejector-based vapor-compression refrigeration cycle was modeled using the data and methods reported in Refs. [1] and [2]. Those studies investigate ejector operating characteristics and their effect on refrigeration-cycle performance using R1234yf. Their data and modeling approach therefore served as the basis for the cycle model developed in this project.

REFPROP is used in this project to evaluate the thermodynamic properties required for simulation of the ejector-based vapor-compression refrigeration cycle.

The refrigeration cycle was simulated using thermodynamic properties obtained from REFPROP together with the ejector model.

After model development and simulation, an optimization stage was carried out to determine the most favorable ejector-cycle operating conditions. Ref. [10] was used as a principal source in this stage. The optimization procedure included the following steps:

The optimization results showed that the ejector can effectively improve the modeled refrigeration-cycle performance. The optimized COP represents a substantial performance level within the operating range investigated.

References

4-1 Scientific Articles Used:

[1] H. Li, F. Cao, X. Bu, L. Wang, and X. Wang, "Performance characteristics of R1234yf ejector-expansion refrigeration cycle," *Applied Energy*, vol. 121, pp. 96-103, 2014.

[2] H. Nishijima, K. Tsuchii, and M. Nakagawa, "Experimental study on the behavior of the two phase flow shock waves occurring in the ejector refrigeration cycle," *Bulletin of the JSME*, vol. 3, no. 6, 2016.

[3] U. Atmaca, A. Erek, and O. Ekren, "Preliminary design of the two-phase ejector under constant area mixing assumption for 5 kW experimental system," *E3S Web of Conferences*, vol. 103, art. no. 01002, 2019.

[4] K. Ameur, Z. Aidoun, and M. Ouzzane, "Modeling and numerical approach for the design and operation of two-phase ejectors," *Applied Thermal Engineering*, 2014. DOI: 10.1016/j.applthermaleng.2014.11.022.

[5] H. K. Ersoy, "Performance improvement of the vapour compression refrigeration cycle by a two-phase constant area ejector," *International Journal of Energy Research*, April 2009. DOI: 10.1002/er.1488.

[6] J. Navarro-Esbrí, R. Cabello, and E. Torrella, "Experimental evaluation of the internal heat exchanger influence on a vapour compression plant energy efficiency working with R22, R134a and R407C," *University Jaume I*, August 2003.

[7] M. Direk, A. Keleşoğlu, and A. Akın, "Theoretical performance analysis of an R1234yf refrigeration cycle based on the effectiveness of internal heat exchanger," *Hittite Journal of Science and Engineering*, vol. 4, no. 1, pp. 23-30, 2017. DOI: 10.17350/HJSE19030000044.

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[9] H. K. Ersoy, and N. Bilir Sag, "Preliminary experimental results on the R134a refrigeration system using a two-phase ejector as an expander," *International Journal of Refrigeration*, 2014. DOI: 10.1016/j.ijrefrig.2014.04.006.

[10] "Parametric study of a vapor compression refrigeration cycle using a two-phase constant area ejector," *World Academy of Science, Engineering and Technology, International Journal of Mechanical, Aerospace, Industrial, Mechatronic and Manufacturing Engineering*, vol. 7, no. 8, 2013.

[11] X. Wang, S. Xu, and C. Xing, "Numerical and experimental investigation on an ejector designed for an 80 kW polymer electrolyte membrane fuel cell stack," Journal of Power Sources, vol. 1, 2024, Available at: www.elsevier.com/locate/jpowsour.

a. Books Used

[12] Moghari, Mojtaba, and Mohammad-Ali Akhavan Behabadi. Design of Refrigeration and Cold-Storage Systems. Tehran: University of Tehran, 2020. 5th ed.

Appendices

4-2 Appendix A

REFPROP Code for Thermodynamic Analysis of the Two-Phase Ejector-Based Vapor-Compression Refrigeration Cycle without a Heat Recuperator for Calculating COP under Different Operating Conditions

```
   clear;
close all;
clc;

syms x_6 q_6;
T_c = 40 + 273.15;
T_e = 5 + 273.15;
eta_mn = 0.85;
eta_sn = 0.85;
eta_d = 0.85;
eta_ms = 0.95;
%delta_p = 14;
w = 0.1;
COP_ejec=zeros(1,100);
delta_p_ejec=0.5:0.5:50;
P_6_ejec=zeros(1,100);
w_ejec=zeros(1,100);
VCC_ejec=zeros(1,100);
i=1;
for delta_p=0.5:0.5:50
    step=0.1;
    while true
        %Condensor
        P_c = refpropm('P','T',T_c,'Q',0,'R1234yf');
        P_3=P_c;
        h_3 = refpropm('H','T',T_c,'Q',0,'R1234yf');
        s_3 = refpropm('S','T',T_c,'Q',0,'R1234yf');
        P_2=P_3;

        %Evaporator
        P_e = refpropm('P','T',T_e,'Q',1,'R1234yf');
        P_9=P_e;
        h_9 = refpropm('H','T',T_e,'Q',1,'R1234yf');
        s_9 = refpropm('S','T',T_e,'Q',1,'R1234yf');

        %Motive Nozzle Outlet
        P_4 = P_9 - delta_p;
        h_4s = refpropm('H','P',P_4,'S',s_3,'R1234yf');
        h_4 = h_3 - eta_mn * (h_3 - h_4s);
        c_4 = sqrt(2 * (h_3 - h_4));
        rho_4 = refpropm('D','H',h_4,'P',P_4,'R1234yf');
        A_4 = 1 / ((1 + w) * rho_4 * c_4);

        %Suction Nozzle Outlet
        P_10 = P_9 - delta_p;
        h_10s = refpropm('H','P',P_10,'S',s_9,'R1234yf');
        h_10 = h_9 - eta_sn * (h_9 - h_10s);
        c_10 = sqrt(2 * (h_9 - h_10s));
        rho_10 = refpropm('D','H',h_10,'P',P_10,'R1234yf');
        A_10 = w / ((1 + w) * rho_10 * c_10);

        %Mixing Section
```

```
        P_5 = P_4;
        c_5 = sqrt(eta_ms) * (1 / (1 + w) * c_4 + w / (1 + w) * c_10);
        h_5 = 1 / (1 + w) * (h_4 + c_4^2 / 2) + w / (1 + w) * (h_10 + c_10^2 / 2) - c_5^2 / 2;
        s_5 = refpropm('S','H',h_5,'P',P_5,'R1234yf');

        %Diffuser Outlet
        h_6 = h_5 + c_5^2 / 2;
        h_6s = h_5 + eta_d * (h_6 - h_5);
        P_6 = refpropm('P','H',h_6s,'S',s_5,'R1234yf');
        x_6 = refpropm('Q','H',h_6,'P',P_6,'R1234yf');

        %Convergence Condition
        q_6 = 1 / (1 + w);
        if abs(x_6-q_6)<0.0001
            break
        elseif x_6<q_6
            w=w+step;
        else
            w=w-step;
            step=step/10;
        end
    end
    AR = (A_4 + A_10) / A_4;
    PLR = P_6 / P_9;

    %gas-liquid separator outlets
    h_1 = refpropm('H','P',P_6,'Q',1,'R1234yf');
    h_7 = refpropm('H','P',P_6,'Q',0,'R1234yf');
    s_1 = refpropm('S','H',h_1,'P',P_6,'R1234yf');
    v_1 =1/(refpropm('D','H',h_1,'P',P_6,'R1234yf'));

    %Compressor Outlet
    eta_comp = 0.874 - 0.0135 * P_c / P_e;
    h_2s = refpropm('H','P',P_2,'S',s_1,'R1234yf');
    h_2 = h_1 + (h_2s - h_1) / eta_comp;

    %Throttle valve
    h_8 = h_7;

    %Outputs
    W_comp = (h_2 - h_1) / (1 + w);
    Q_evap = w * (h_9 - h_8) / (1 + w);
    COP = Q_evap / W_comp;
    VCC = w * (h_9 - h_8) / v_1;
    COP_ejec(i)=COP;
    w_ejec(i)=w;
    VCC_ejec(i)=VCC;
    P_6_ejec(i)=P_6;


    i=i+1;
end

figure
plot(delta_p_ejec,COP_ejec,'b-*')
grid on
title('\DeltaP - COP Diagram','fontsize',14)
```

```
xlabel('\DeltaP(kPa)','fontsize',14)
ylabel('COP','fontsize',14)

figure
plot(delta_p_ejec,VCC_ejec,'g-*')
grid on
title('\DeltaP - VCC Diagram','fontsize',14)
xlabel('\DeltaP(kPa)','fontsize',14)
ylabel('VCC','fontsize',14)
```

4-3 Appendix B

REFPROP Code for Thermodynamic Analysis of the Two-Phase Ejector-Based Vapor-Compression Refrigeration Cycle with a Heat Recuperator

```
clc;

clear;

format long
syms x_6 q_6;
%T_c = 40 + 273.15;
T_e = 5 + 273.15;
eta_mn = 0.9;
eta_sn = 0.9;
eta_d = 0.8;
eta_ms = 1;
eta_he=0.8;
delta_p = 14;
mo = 0.1;
P_6_v=zeros(1,100);
mo_v=zeros(1,100);
VCC_v=zeros(1,100);
i=1;
for T_c=45+273.15:1:65+273.15
    T_7_a=278;
    step_m=0.1;
    step_T=0.1;
    while true
        %condensor
        P_c = refpropm('P','T',T_c,'Q',0,'R1234yf');
        P_3=P_c;
        h_3 = refpropm('H','T',T_c,'Q',0,'R1234yf');
        s_3 = refpropm('S','T',T_c,'Q',0,'R1234yf');
        %evaperator
        T_9=T_e;
        P_e = refpropm('P','T',T_e,'Q',1,'R1234yf');
        P_9=P_e;
        h_9 = refpropm('H','T',T_e,'Q',1,'R1234yf');
        s_9 = refpropm('S','T',T_e,'Q',1,'R1234yf');
        %heat exchanger
        T_7_p=T_7_a-eta_he*(T_7_a-T_9);
        h_7_a = refpropm('H','T',T_7_a,'Q',0,'R1234yf');
        P_7_a=refpropm('P','T',T_7_a,'Q',0,'R1234yf');
        P_7_p=P_7_a;
        P_9_p=P_9;
        h_7_p=refpropm('H','T',T_7_p,'P',P_7_p,'R1234yf');
        h_9_p=h_7_a+h_9-h_7_p;
        s_9_p = refpropm('S','P',P_9_p,'H',h_9_p,'R1234yf');

        P_4 = P_9 - delta_p;
        h_4s = refpropm('H','P',P_4,'S',s_3,'R1234yf');
        h_4 = h_3 - eta_mn * (h_3 - h_4s);
        c_4 = sqrt(2 * (h_3 - h_4));
        rho_4 = refpropm('D','H',h_4,'P',P_4,'R1234yf');
        A_4 = 1 / ((1 + mo) * rho_4 * c_4);

        P_10 = P_9_p - delta_p;
        h_10s = refpropm('H','P',P_10,'S',s_9_p,'R1234yf');
        h_10 = h_9 - eta_sn * (h_9_p - h_10s);
        c_10 = sqrt(2 * (h_9_p - h_10s));
        rho_10 = refpropm('D','H',h_10,'P',P_10,'R1234yf');
        A_10 = mo / ((1 + mo) * rho_10 * c_10);

        P_5 = P_4;
        c_5 = sqrt(eta_ms) * (1 / (1 + mo) * c_4 + mo / (1 + mo) * c_10);
```

```
            h_5 = 1 / (1 + mo) * (h_4 + c_4^2 / 2) + mo / (1 + mo) * (h_10 + c_10^2 / 2)
- c_5^2 / 2;
            s_5 = refpropm('S','H',h_5,'P',P_5,'R1234yf');

            h_6 = h_5 + c_5^2 / 2;
            h_6s = h_5 + eta_d * (h_6 - h_5);
            P_6 = refpropm('P','H',h_6s,'S',s_5,'R1234yf');
            x_6 = refpropm('Q','H',h_6,'P',P_6,'R1234yf');
            q_6 = 1 / (1 + mo);

            T_7 = refpropm('T','P',P_6,'Q',0,'R1234yf');
            if abs(x_6-q_6)<0.0001
                if abs(T_7-T_7_a)<0.0001
                    break
                elseif T_7>T_7_a
                    T_7_a=T_7_a+step_T;
                else
                    T_7_a=T_7_a-step_T;
                    step_T=step_T/10;
                end
            elseif x_6<q_6
                if abs(T_7-T_7_a)<0.0001
                    mo=mo+step_m;
                elseif T_7>T_7_a
                    mo=mo+step_m;
                    T_7_a=T_7_a+step_T;
                else
                    mo=mo+step_m;
                    T_7_a=T_7_a-step_T;
                    step_T=step_T/10;
                end
            else
                if abs(T_7-T_7_a)<0.0001
                    mo=mo-step_m;
                    step_m=step_m/10;
                elseif T_7>T_7_a
                    mo=mo-step_m;
                    step_m=step_m/10;
                    T_7_a=T_7_a+step_T;
                else
                    mo=mo-step_m;
                    step_m=step_m/10;
                    T_7_a=T_7_a-step_T;
                    step_T=step_T/10;
                end
            end
        end
        AR = (A_4 + A_10) / A_4;
        PLR = P_6 / P_9;
        h_1 = refpropm('H','P',P_6,'Q',1,'R1234yf');
        h_7 = refpropm('H','P',P_6,'Q',0,'R1234yf');
        T_7 = refpropm('T','P',P_6,'Q',0,'R1234yf');

        P_2=P_3;
        s_1 = refpropm('S','H',h_1,'P',P_6,'R1234yf');
        eta_comp = 0.874 - 0.0135 * P_c / P_e;
        h_2s = refpropm('H','P',P_2,'S',s_1,'R1234yf');
        h_2 = h_1 + (h_2s - h_1) / eta_comp;
        h_8 = h_7_p;
        mo;
        P_6;
        v_1 =1/(refpropm('D','H',h_1,'P',P_6,'R1234yf'));
        W_comp = (h_2 - h_1) / (1 + mo);
        Q_evap = mo * (h_9 - h_8) / (1 + mo);
        COP = Q_evap / W_comp;
        VCC = mo * (h_9 - h_8) / v_1;
        COP_v(i)=COP;
```

```
        mo_v(i)=mo;
        VCC_v(i)=VCC;
        P_6_v(i)=P_6;
        T_7_v(i)=T_7;
        i=i+1;
    end
    for i=1:1:21
        COP(i)=COP_v(i);
    end
    T_c_v=45:1:65;
    figure
    plot(T_c_v,COP,'g-*')
    hold on
```

4-4 Appendix C

Machine-Learning Algorithm Code for COP Prediction

```
function [trainedModel, validationRMSE] = trainRegressionModel(trainingData)
inputTable = trainingData;
predictorNames = {'delta_p', 'T_C', 'T_E'};
predictors = inputTable(:, predictorNames);
response = inputTable.COP;
isCategoricalPredictor = [false, false, false];

% Train a regression model
% This code specifies all the model options and trains the model.
regressionTree = fitrtree(...
    predictors, ...
    response, ...
    'MinLeafSize', 4, ...
    'Surrogate', 'off');

predictorExtractionFcn = @(t) t(:, predictorNames);
treePredictFcn = @(x) predict(regressionTree, x);
trainedModel.predictFcn = @(x) treePredictFcn(predictorExtractionFcn(x));

trainedModel.RequiredVariables = {'T_C', 'T_E', 'delta_p'};
trainedModel.RegressionTree = regressionTree;
trainedModel.About = 'This struct is a trained model exported from Regression Learner R2021a.';

inputTable = trainingData;
predictorNames = {'delta_p', 'T_C', 'T_E'};
predictors = inputTable(:, predictorNames);
response = inputTable.COP;
isCategoricalPredictor = [false, false, false];

partitionedModel = crossval(trainedModel.RegressionTree, 'KFold', 5);

validationPredictions = kfoldPredict(partitionedModel);

validationRMSE = sqrt(kfoldLoss(partitionedModel, 'LossFun', 'mse'));
```

Code for Calling the Implemented Prediction Function

```
close all;
clc;

data = readtable('Book1.xlsx');

trainingData = data;

[trainedModel, validationRMSE] = trainRegressionModel(trainingData);

disp(['Validation RMSE: ', num2str(validationRMSE)]);
delta_p_new = [35; 33];
T_C_new = [303.15; 320.15];
T_E_new = [279.15; 279.65];
newData = table(delta_p_new, T_C_new, T_E_new, 'VariableNames', {'delta_p', 'T_C',
    'T_E'});
predictedCOP = trainedModel.predictFcn(newData);

disp('Predicted COP values:');
disp(predictedCOP);
```

4-5 Appendix D

Optimization-Algorithm Code Used to Determine the Best COP Operating Condition

```
close all;
clc;

data = readtable('Book1.xlsx');
trainingData = data;
[trainedModel, validationRMSE] = trainRegressionModel(trainingData);

disp(['Validation RMSE: ', num2str(validationRMSE)]);


lb = [0.5, 30+273.15, -5+273.15];
ub = [50, 50+273.15, +5+273.15];

nvars = 3;

options = optimoptions('ga', 'Display', 'iter');
[optimalX, maxCOPNeg] = ga(@(x) objectiveFunction(x, trainedModel), nvars, [], [],
[], [], lb, ub, [], options);

maxCOP = -maxCOPNeg;

disp('Optimal values:');
disp(['delta_p: ', num2str(optimalX(1))]);
disp(['T_C: ', num2str(optimalX(2))]);
disp(['T_E: ', num2str(optimalX(3))]);
disp(['Max COP: ', num2str(maxCOP)]);


function negCOP = objectiveFunction(x, trainedModel)
    newData = table(x(1), x(2), x(3), 'VariableNames', {'delta_p', 'T_C', 'T_E'});
    predictedCOP = trainedModel.predictFcn(newData);
    negCOP = -predictedCOP;
end
```